READ ME FIRST —
Summary of the Research So Far

topo-pilot: topology of a latent world in transformer representations

The questions, the methods, every experiment and stage to date (E0–E4, X4, X5, the oracle profile, K4, K4b, C1′, D-1, D-2a, X2a, X6 stage 0 and X6 stage 1), the results including the negative and inconclusive ones, and what remains open

21 October 2026

Living document; updated as runs complete.

Compiled from the repository (`topo-pilot`, branch `main`), the coordination mailbox (`agent-handoff`) and the execution stores beside it. Every number is taken from a cited file. X6 stage 1 reached its preregistered decision on 21 October 2026; its decision-stage report is cited where its numbers are used. This text changes no decision. Where it summarises, the cited reviews, reports and frozen protocols are authoritative.

Abstract

I ask whether a small causal transformer, trained from scratch only to predict the next symbol of sequences emitted by a hidden random walk on a world of known topology, develops internal structure that reflects that topology, where such structure comes from, and whether the model uses it. The pilot compares a 12×12 cylinder with a 12×12 square under matched dynamics and emissions, with an exact Bayesian oracle, matched emission controls, and representation baselines, and decides one primary hypothesis by a rule frozen before any model was trained. Under the smooth emission, training produces a persistent-H_1 signal in the final residual stream that separates the cylinder from the square in 24 of 24 panels across three seeds (median contrast 0.158–0.176 against a noise envelope of 0.084–0.097), and its leading H_1 class aligns with the world's periodic coordinate (agreement 0.88–0.89, exact-belief ceiling 0.92). The activation geometry tracks the 32-symbol Bayesian belief more than the last-symbol posterior. A timing hypothesis was rejected, the matched permuted-emission control turned out to be one-sided, the last-symbol embedding alone carries 0.47–0.89 of the contrast, linear probes recover no more belief than the exact next-symbol law does, and a clean-rerun gate failed on one missing historical record. More than a dozen follow-up studies, almost all preregistered, then went after the origin of the circular structure, its causal use, and the necessity of topological distinctions for longer prediction horizons. Most stopped at preregistered feasibility gates (INCONCLUSIVE) or returned scoped negatives; positive results are oracle facts about designed processes (D-1, X2a, X6 stage 0). In the most direct origin test (K4b), the covariance of the learned chart phase with the part of the belief that the last four observations do not determine was excluded from beyond a ± 0.01 margin in all three models; the registered margin, however, lies above what the oracle's own phase achieves. The current study, X6 stage 1, trains twelve models on a non-abelian (S_3) transport process to test whether trained transformers use the order of seam crossings beyond every commutative summary; all validity gates passed and the preregistered outcome is INCONCLUSIVE (rule 9). In every seed the trained models carry a small but clearly non-zero order-dependent component on natural histories (mean $\tilde{\beta} = 0.061$, lower bound 0.0595; commutative controls and initialisations near zero), an order of magnitude below the registered effect size of 0.25, and they do not beat the designed W2 bar; one commutative control's designed placebo response prevents the registered ORDER_NOT_DETECTED verdict. No causal claim about any trained model has been established.

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1 Introduction

Large language models are trained to predict the next token, yet they are often described as carrying "world models". Whether, and in what sense, next-token training produces representations that mirror the latent structure of the process generating the data is an empirical question that is hard to answer on natural text, because the latent structure is unknown. Synthetic processes whose latent state, dynamics and exact Bayesian predictor are fully known make the question tractable. Prior work established that transformers trained on hidden Markov models (HMMs) linearly represent the geometry of the Bayesian belief state in their residual stream [67, 75], and that sequence models trained on games or walks can carry recoverable world state [14, 48, 80]. Topology—global invariants such as the number of independent loops—had not been controlled in this setting.

The **topo-pilot** project trains a small causal transformer from scratch on symbols emitted by a lazy random walk on one of two worlds with the same number of states and the same local rule: a *cylinder* (one independent loop, $\beta_1 = 1$) and a *square* ($\beta_1 = 0$). It asks three nested questions, fixed in the binding specification `docs/RESEARCH_SUMMARY.md` §1 and later in the owner's research charter (`agent-handoff/NEXT_RESEARCH_AND_WORKFLOW_EN.md`):

RQ1 Existence. Does the context-dependent representation acquire structure that reflects the latent topology?

RQ2 Origin. If so, does that structure come from the geometry of the observations, from beliefs accumulated over the history, or only from what is needed to predict the next token? (Charter priority 1: how much of the circular structure is explained by the last symbol or a short context, and how much requires long history?)

RQ3 Use. Does the model use the structure for prediction, or is it merely detectable? (Charter priority 2.) A third charter priority asks whether some topological distinctions are unnecessary for the next symbol but necessary for a longer future.

Two secondary programmes extend the question: X5 (does geometry change learning speed?) and X6 (topology richer than H_1: non-abelian transport, where the *order* in which loops are traversed matters).

The project is also an experiment in research process. I froze every decision rule before the data it decides existed. Each protocol was planned twice independently and reviewed result-blind before freezing, and the critical tests were written by authors who never saw the implementation. Compute is metered, fresh test histories are protected by exposure registries, and failures are classified as technical (correctable at most twice), inconclusive, or scientific. I did much of the work with language-model agents: OpenAI Codex and Anthropic Claude models served as independent planners, result-blind reviewers, implementers and test authors, and a Claude session acted as coordinator (issuing rulings and writing reviews under my authority) with workers launched per job, all communicating through a version-controlled mailbox. I approved every scientific decision point.

Summary of findings. Table 1 lists the main outcomes; Figure 1 places all of them in time.

1. **RQ1, positive under the smooth emission O.** The preregistered F1 rule is supported in every panel and replicate; the detected loop is the world's periodic circle (§5).
2. **RQ2, partially answered.** The geometry follows the full-history belief, also among histories with the same last symbol, and the learned circle exceeds an optimistic last-symbol bound. But the last-symbol embedding already carries much of the contrast, the timing hypothesis was rejected, beliefs converge quickly with history length, and the most direct test (K4b) found the beyond-four-symbol covariance of the learned phase to be smaller than

Table 1: Headline outcomes. "Label" is the outcome vocabulary of the governing frozen protocol. Sources: `README.md`; `reports/E2--E4`; `state/DECISIONS.md`; `agent-handoff/jobs/<job>/review.md`.

Study	Label	One-line reading
F1 (pilot, condition O)	SUPPORTED	$D > 0$ in 24/24 panels; median above envelope in 3/3 seeds
H2 (pilot)	sign-consistent	geometry tracks $b^{(32)}$ over $b^{(1)}$ in 48/48 panels
H3 (F1 rule under P)	REJECTED, uninformative	even the exact belief fails under P
H3′ (belief alignment, P)	SUPPORTED	ρ_s 0.06–0.11 $\rightarrow$ 0.70, beats mixing baselines
H4 (timing)	REJECTED	S rises at step 50, history use at step 100
G-H (clean rerun)	FAIL	307/307 shared artifacts bitwise; one historical record missing
X4 Phase B / B2	INCONCLUSIVE / scoped NO-GO	no horizon-1/horizon-2 witnesses at registered resolution
X4-F (causal)	INCONCLUSIVE	linear eraser leaves nonlinear information; no matched family
X5 G1	H5 REJECTED	paired delays 250/0/0 steps (kernel confounded with speed)
K8 / K4	headroom FAIL / INCONCLUSIVE	no room at $K = 8$; chart gate failed at $K = 4$
D-1 (oracle)	PASS	orientability unnecessary at horizon 1, necessary at horizon 2
X2a stage 0 (oracle)	GO_P1_ NONADDITIVE	component identity needs long, non-additive history
D-2a (trained)	NOT_LEARNED	both gate models failed the band-edge criterion
C1′ (causal)	INCONCLUSIVE/ FAMILY_UNMATCHED	intervention moved protected readouts
K4b (origin)	EXCLUSION	$\lvert\Delta_4\rvert < 0.01$ in 3/3 models; oracle comparator 0.0085
X6 stage 0 (oracle)	GO_X6_NATURAL	order effect certified beyond Parikh statistics
X6 stage 1 (trained)	INCONCLUSIVE (rule 9)	all validity gates PASS; mean $\tilde{\beta} = 0.061$, below the registered 0.25

0.01—a margin that the oracle's own phase also does not reach (§6).

3. **RQ3, open.** Both intervention designs (X4-F and C1′) stopped at preregistered feasibility gates before any causal endpoint was scored (§7).

4. **Horizons and richer topology.** Exact oracle studies prove the existence of an orientability distinction that is invisible at horizon 1 and needed at horizon 2 (D-1), but a trained-model study of it failed its learnability gate (D-2a). An S_3 transport process was certified to carry order information beyond every commutative statistic (X6 stage 0); the trained follow-up (X6 stage 1) reached its preregistered decision, INCONCLUSIVE: a weak but reproducible natural order sensitivity, far below the registered size (§8, §10, §10.2.1).

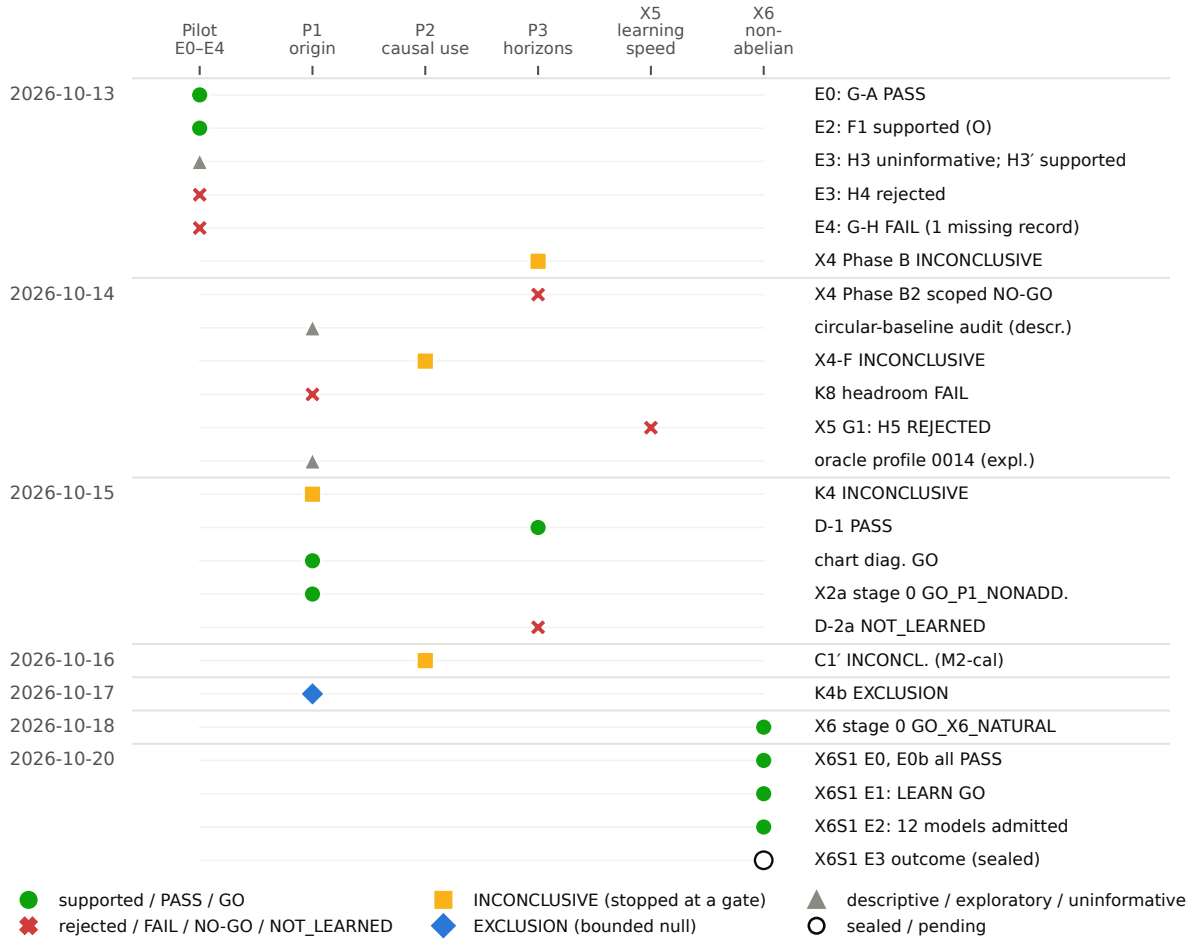


Figure 1: Programme timeline. Each row is one recorded stage outcome, placed in the column of the research question it served (P1–P3 are the owner charter's priorities). Order within a day follows the decision log; spacing is not to scale. Sources (one per row, in `make_figures.py`): `state/DECISIONS.md` D-017, D-049, D-052, D-054, D-057, D-063, D-073, D-074; `reports/E2/summary.md`, `reports/E3/summary.md`; `x4-history-feas-store/GATE_REPORT.json`; `agent-handoff/jobs/<job>/review.md` for jobs 0018, 0020, 0023, 0025, 0033, 0044, 0057, 0066, 0089, 0092 and 0093.

2 Background and related work

2.1 Transformers, hidden Markov models and belief states

An HMM with transition matrix T and emission matrix E induces, for every observed history, a posterior over latent states: the belief state, obtained by Bayesian filtering [69]. The set of reachable beliefs, together with the map from belief to future-symbol distributions, is the object studied by computational mechanics, where the minimal sufficient statistic of the past for the future (the causal state, or ε-machine state) is the natural notion of what an optimal predictor must remember [18, 76]. Predictive-state and observable-operator representations express the same information through future-observation statistics [42, 78]. Shai et al. showed that transformers trained on small edge-emitting HMMs linearly encode the belief-state simplex, including fractal belief geometries, in the residual stream, sometimes only in the concatenation of layers [75]; Piotrowski et al. explained the layer-wise geometry as constrained belief updates by attention followed by nonlinear correction [67]. Related accounts treat in-context learning as implicit Bayesian inference over latent concepts [82], show how transformers learn Markov-chain statistics in context through induction-like heads with stage-wise dynamics [9, 21, 60], and study what function classes can be learned in context

[30]. More recent preprints report factored representations when latent factors are independent [74] and belief geometry in pretrained models in context [3].

Relation to this project. The pilot uses exactly this oracle machinery (§4.1): the exact filter supplies the 32-symbol belief $b^{(32)}$, the last-symbol posterior $b^{(1)}$ and the future laws F_1, F_2, and every learned geometry is compared with these model-free spaces on the same histories. Two design choices come directly from this literature. First, the axis-wise emission permutation P (§4.1) preserves the exact product structure of the smooth emission, because factor-wise representations are expected [74]. Second, layer-wise and concatenated hooks are analysed because belief geometry need not sit in one layer [67, 75]. The pilot's addition is a controlled *topological* contrast at matched state count, and persistent homology on clouds of individual histories rather than on per-state centroids.

2.2 World models, emergent geometry and in-context representations

Othello-playing sequence models carry recoverable board state [48]; models trained on random walks can predict well without a coherent world model by stricter metrics [80]; pretrained language models represent space and time linearly [33] and some features are irreducibly multi-dimensional, including circular ones such as days of the week [26]. Pretrained models reorganise their representations to match a graph defined in context, but the analysis there uses per-node averages, whose principal components reproduce the spectral embedding of the graph [65]. Closest to the pilot in spirit, Jerpelea planted a ring-structured latent Markov variable in generated text and found ring geometry in a from-scratch student, using centroids and PCA [43]; Brenner et al. studied fully observed lattice walkers, whose residual stream encodes running averages of steps [14].

Relation. These results make two pitfalls concrete. Per-state averages import the world's structure from outside the model, and short-range symbol mixing alone can create global lattice structure. The pilot therefore forbids centroids, UMAP/t-SNE and k-NN geodesics as evidence, and adds two model-free mixing baselines (R2, previous-token mixing of a spectral symbol embedding; R3, context average), together with the trained last-symbol embedding (R1) (`docs/RESEARCH_SUMMARY.md` §7.2, §7.6). The claim of the pilot is accordingly narrow: the general phenomenon "periodic latent dynamics can shape periodic geometry" has prior art [43]; the pilot contributes a matched two-topology comparison with these controls. I do not claim priority.

2.3 Formal languages, automata, groups and state tracking

Theoretical work bounds what fixed-depth self-attention can compute: hard and soft attention cannot robustly recognise parity or Dyck languages at arbitrary length [34], empirical studies map the formal languages transformers learn [8], and a Chomsky-hierarchy benchmark finds transformers weak at state tracking beyond regular languages without structural help [20]. Log-precision transformers lie in uniform TC^0 [55]; since the word problem of S_5 is NC^1-complete by Barrington's theorem [4], neither transformers nor common state-space models can track general non-solvable group state at arbitrary length unless $\mathrm{TC}^0 = \mathrm{NC}^1$ [56, 73]; negative eigenvalues restore some state tracking in linear RNNs [32]. For *solvable* groups, transformers learn shallow "shortcuts" whose existence follows from the Krohn–Rhodes decomposition [46, 49]. Empirical mechanistic work shows that networks learn group operations through representation-theoretic circuits [16], that language models track permutation state with associative and parity-associative mechanisms [47], and a recent preprint argues that standard transformers learn cosets of normal subgroups [85].

Relation. X6 rests on this distinction. Its transport group S_3 is non-abelian but solvable, so the composition task is expressible at fixed depth [49], so expressivity is not in doubt; the question is whether next-token training on a *topologically generated, implicitly observed* group word yields

order-sensitive predictions beyond commutative summaries. The commutative competitors of X6 form a ladder (parity, which is the quotient S_3/A_3; signed crossing counts, i.e. the H_1 class; unsigned counts; the full Parikh vector of directed crossings, the strongest commutative statistic of a word [64]). The coset account predicts either full order use or parity-level behaviour for S_3, whose normal subgroups are trivial, A_3 and S_3 [85]; X6 records this as a named competing hypothesis, checked against the preprint's abstract only.

2.4 Order sensitivity versus bag-of-words

Transformer language models use much less word-order information than their architecture permits: ablating order within long-range context changes little [62], and pretraining on shuffled text retains much downstream performance [79]. *Relation.* X6 makes the order question exact: on designed history families with identical token and directed-bigram multisets, the oracle's next-symbol law still differs, so any bag-of-tokens predictor has a certified excess loss (§10.1). On the natural distribution, however, bag mechanisms are *not* excluded by the stage-1 design, and the protocol says so.

2.5 Mechanistic interpretability, probing and interventions

Circuit-level analysis of transformers [24], induction heads [63] and superposition [25] supply the vocabulary of residual stream, heads and readouts used here. Linear probes [1] measure decodability, not use; control tasks and selectivity are needed to interpret them [6, 39]. Linear concept erasure [7, 70] removes linearly decodable information but may leave nonlinearly decodable information; causal claims require interventions such as interchange interventions, causal abstraction or activation patching [31, 37, 54, 84].

Relation. The pilot's probes compare decoding of $b^{(32)}$ from activations against decoding from the exact next-symbol law and from a one-hot last symbol, which acts as a competitor-matched control in the spirit of Hewitt and Liang [39]. Both of the project's causal designs met the known failure modes of interventions: X4-F's targeted linear eraser removed linear recoverability while a nonlinear (RBF) decoder still recovered the coordinate at 0.998, and no random eraser matched its damage; C1′'s state-forcing direction shifted protected readouts that matched controls left nearly untouched (§7).

2.6 Grokking and learning dynamics

Small transformers on algorithmic tasks can generalise long after fitting the training data [68]; mechanistic progress measures reveal a gradual circuit formation behind apparently sudden transitions [59], and different runs can implement different algorithms for the same modular task [86]. *Relation.* H4 asks a timing question of this kind (does topology appear only after history use begins?), and X6 stage 1 trains for 30,000 updates because D-2a's 10,000-update recipe showed a flat dev loss from step 1,000 (`agent-handoff/jobs/0033/review.md`); I chose the longer run; it is not a tuned value.

2.7 Topological data analysis of representations

Persistent homology tracks homology classes across a filtration [15, 22, 88]; diagrams are stable under perturbation [17], admit confidence sets [27], and circular coordinates can be extracted from persistent cohomology classes [19]. Vietoris–Rips barcodes are computed efficiently by Ripser [5]; universality of random persistence diagrams is conjectured or proved under specific conditions [10]. Persistent homology has revealed toroidal structure in grid-cell populations [29], tracked topology through deep networks [58, 71], and been applied to language-model activations, where matched

Gaussian clouds produce hundreds of spurious H_1 bars and pseudoreplication must be avoided [52]; another study argues that transformer token clouds within a sentence are topologically simple [28]. Sheaves and local systems formalise local-to-global consistency [35] and have entered neural architectures [11]; local systems and holonomy are classical [36].

Relation. The pilot computes Vietoris–Rips homology over $\mathbb{F}_2$ on clouds of 512 individual histories, normalises the longest finite H_1 bar by a cloud scale, and reports a Gaussian floor as context, never as a threshold. Following Hurlbert [41] and Malhotra et al. [52], panels within a run are not treated as replicates; the decision uses sign consistency across three seeds and eight disjoint panels. The ground truth is computed from an explicit cubical complex, because the transition graph of the square has 121 independent cycles that are filled faces, not holes. X6 uses a local system (a flat S_3 connection with one non-flat face) in the *generator*; nothing is claimed about the trained model being a sheaf.

2.8 Concentration bounds and resampling

The bootstrap [23] is used throughout for descriptive intervals and several confirmatory rules, with stated nominal (not distribution-free) coverage. Where a finite-sample guarantee is needed, X6 stage 1 uses an empirical-Bernstein bound for a bounded loss: variance-adaptive bounds of this type are due to Audibert et al. [2] and Maurer and Pontil [53], in contrast to the variance-blind Hoeffding inequality [40]; see Boucheron et al. [12] for background. The protocol derives its own version with a paired U-statistic for the variance, at a small cost in constants relative to Maurer and Pontil [53] (§10.2).

2.9 Preregistration, reproducibility and leakage

Undisclosed analytic flexibility inflates false positives [77]; preregistration separates confirmatory from exploratory analysis [57, 61]. In machine learning, variance across seeds and tooling undermines single-run claims [13, 38, 87], and reproducibility programmes have been proposed [66]. Data leakage and test contamination are a recurrent source of invalid results [44, 45, 51, 72]. *Relation.* The methods of §4 are meant as a strict response to these problems: frozen rules, result-blind review, three-valued outcomes, exposure registries for fresh test histories, sealed validation quantities, and bitwise rerun checks.

3 The research programme

The programme grew in three phases.

Phase 1: the pilot (E0–E4, 12–13 October 2026). A staged plan (stages S0–S8, exit points E0–E4; `docs/STAGES.md`) built and validated the measurement pipeline on ground truth (E0), established that the exact oracle spaces separate the worlds (E1), decided F1 and H2 (E2), ran the permuted-emission control and the timing analysis (E3), and added controls, probes, circular coordinates, sensitivity analyses and a clean rerun (E4). Four human checkpoints (H1–H4) required owner approval. Seventeen models were trained on CPU.

Phase 2: follow-ups on origin, use and horizons (13–17 October). After E4 I fixed the charter priorities quoted in §1. Each follow-up was a separate, frozen protocol: X4 Phase B/B2 (do horizon-2 distinctions exist that horizon 1 hides?), the circular-baseline audit, X4-F (causal use by erasure), the oracle profile (how fast does the circular posterior converge with suffix length?), the K8 cohort, K4 and K4b (does the learned phase covary with information beyond the last K symbols?), C1 and C1′ (causal use by crossed counterfactual state forcing), D-1 and D-2a (an orientability distinction needed only at horizon 2: oracle proof, then trained models),

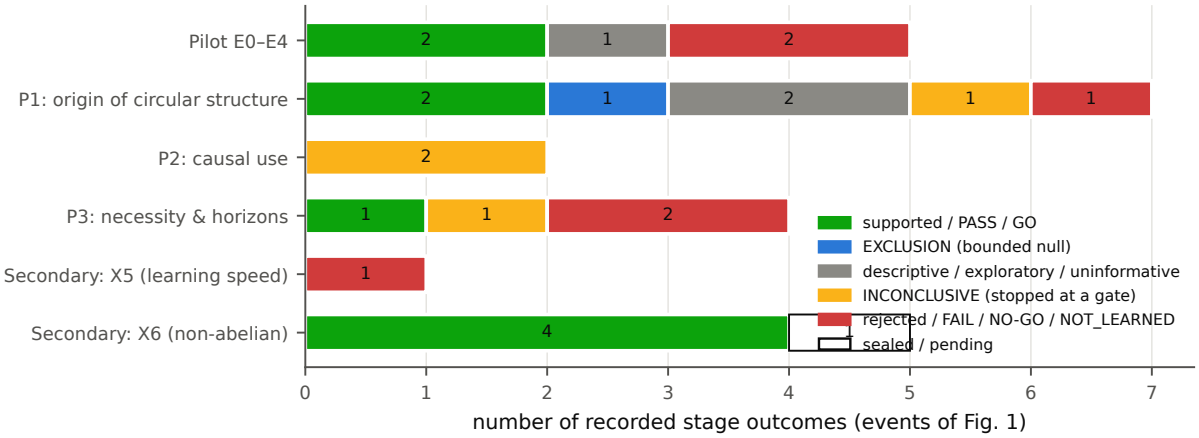


Figure 2: Outcome summary by research question: the number of recorded stage outcomes of Figure 1 in each outcome class. Supported outcomes are concentrated in the pilot and in oracle-only studies; trained follow-ups mostly stopped at gates. Source: the event list in `make_figures.py` (`EVENTS`), each with its cited review or decision record.

X2a stage 0 (component identity beyond short windows), and X5 G1 (learning speed under two kernels).

Phase 3: non-abelian topology (17 October–). X6 stage 0 certified, by exact computation, that an S_3 transport process carries order information beyond every commutative statistic; X6 stage 1 (frozen 18 October, authorised 20 October) trains twelve models on it, on a free cloud GPU, with all evaluation local.

How the questions connect. The pilot answered RQ1 positively under O and left RQ2 split between "observation geometry" and "history" (the R1 limitation). The origin studies sharpened RQ2 into a quantitative question about suffix lengths, which the oracle profile showed can only be asked at small K, because the circular posterior converges quickly. The causal studies went at RQ3 directly and stopped at feasibility gates; they showed that the circular information is encoded nonlinearly and that naive interventions are not specific. The horizon studies moved from a scoped negative on the pilot world (B2) to a constructed world where a topological distinction is provably needed only at horizon 2 (D-1); the trained version (D-2a) did not learn. X2a and X6 stage 0 asked, in the oracle, for processes where long and order-dependent history is genuinely required, so that a trained model's behaviour can discriminate between explanations. X6 stage 1 is the first trained test on such a process.

4 Methodology

4.1 The synthetic processes and the exact oracle

Worlds. A 12×12 lattice with 4-neighbourhood, $N = 144$ states. The square has no identification; the cylinder is periodic in x. Ground-truth Betti numbers are computed from an explicit cubical complex over $\mathbb{F}_2$: square $\beta = (1, 0, 0)$, cylinder $(1, 1, 0)$. The primary metric is the intrinsic flat metric $d_M((x, y), (x', y')) = \sqrt{\min(|x - x'|, 12 - |x - x'|)^2 + (y - y')^2}$ on the cylinder (without the minimum on the square).

Dynamics. A lazy walk normalised by the maximal degree,

$$T_{ij} = \frac{1 - p_{\text{stay}}}{D_{\max}} \ (i \sim j), \qquad T_{ii} = 1 - \deg(i)\,\frac{1 - p_{\text{stay}}}{D_{\max}}, \qquad p_{\text{stay}} = 0.2,\ D_{\max} = 4, \tag{1}$$

which is symmetric and doubly stochastic, so the stationary law π is uniform.

Emissions. Four conditions (`docs/RESEARCH_SUMMARY.md` §4):

- **O** (main): 36 symbols with centres $c_{ab} = (0.5 + 2a, 0.5 + 2b)$ on a 6×6 grid, $E_{zj} \propto \exp(-d_M(z, c_j)^2/2\sigma^2)$, symbol names randomly permuted; $\sigma = 1.25$ was the first candidate to pass the generator-only calibration rule. O factorises exactly as $E^x[x, a]\, E^y[y, b]$.
- **P** (matched control): $E_P[(x, y)] = E_O[(\sigma_x(x), \sigma_y(y))]$ with fixed axis permutations, preserving rows, entropies, symbol marginals and the product structure while destroying spatial smoothness.
- **N** (negative control): $E_{zw} = 1/36$, nothing is learnable about the state.
- **U** (sanity control): one symbol per state ($V = 144$).

Sequences and the exact predictor. $z_1 \sim \pi$; for $t = 1, \ldots, 33$ emit $w_t \sim E[z_t]$ and transition. The model sees $[\mathrm{BOS}, w_1, \ldots, w_{32}]$ and predicts w_{t+1} at every position. The filter, in row vectors and float64, is

$$a_1 = \pi, \qquad b_t = \frac{a_t \odot E_{:,w_t}}{a_t E_{:,w_t}}, \qquad a_{t+1} = b_t T, \qquad q_t = b_t T E. \tag{2}$$

From it come $b^{(32)} = b_{32}$, the last-symbol posterior $b^{(1)} = r_{w_{32}}$ with $r_w = \pi \odot E_{:,w}/(\pi E_{:,w})$, the next-symbol law $F_1 = bTE$ and the two-symbol law $F_2[i, j] = b\, T\, \mathrm{diag}(E_{:,i})\, T\, E_{:,j}$. The *context gain* measures what the full history adds to the last symbol, and the *learning gate* measures how much of it the model captures:

$$G_C = \overline{\mathrm{KL}(q_C \,\|\, q_{\mathrm{last}})}, \qquad \mathrm{regret} = \overline{\mathrm{KL}(q_C \,\|\, p_{\mathrm{model}})}, \qquad \mathrm{captured} = 1 - \mathrm{regret}/G_C \;\geq\; 0.8. \tag{3}$$

Later studies reuse the same filter and sampler on other worlds: the 4×3 cylinder/Möbius seam band with a hidden world bit (D-1, D-2a), two 12×12 tori with a Dehn-twisted emission (X2a), and a 4×4 torus carrying an S_3 local system (X6).

4.2 The transformer learners

Every trained model in the project uses the same pilot architecture (`docs/RESEARCH_SUMMARY.md` §6): a decoder-only pre-LN transformer [81, 83] with 4 blocks, $d = 128$, 4 heads, MLP width 512, GELU, learned positional embeddings, final LayerNorm, untied unembedding, no dropout, about 0.8M parameters (802,304 in X6, with 18 output symbols). Training uses AdamW [50] with learning rate 10^{-3}, 200 warm-up steps, cosine decay, weight decay 0.1, clipping 1.0 and batch 128. The pilot trained for 5,000 updates on 131,072 trajectories per replicate, with checkpoints at 0, 50, 100, 200, 400, 700, 1,000, 1,500, 2,000, 3,000, 4,000 and 5,000; seeds 1729, 2718 and 3141 change data, initialisation, symbol names and batch order together. The primary hook is `resid_post_3`, the residual stream after the last block and before the final LayerNorm. D-2a trained 10,000 updates (with optional direct joint heads for 2- and 3-symbol futures), and X6 stage 1 trains 30,000 updates on 2^{20} histories per model.

4.3 Measurement: persistent homology and the F1 rule

For each replicate, eight disjoint panels of 512 test histories (one per trajectory, at position 32) are shared by every hook, checkpoint and space. On each cloud a Vietoris–Rips filtration over $\mathbb{F}_2$ in dimensions 0–1 is computed, and

$$S = \max_{\text{finite } H_1 \text{ bars}} \frac{\text{death} - \text{birth}}{s}, \qquad s = q_{95}\{d(x, x') > 0\}. \tag{4}$$

For replicate r and panel i, the contrast removes what the architecture produces at initialisation and what training produces in either world:

$$D_{r,i} = (S^{\text{cyl}}_{\text{final},i} - S^{\text{cyl}}_{\text{init},i}) - (S^{\text{sq}}_{\text{final},i} - S^{\text{sq}}_{\text{init},i}), \qquad u_r = \max_c q_{95}\{|S_{c,i} - S_{c,j}| : i < j\}, \tag{5}$$

with c ranging over the four cells. **F1 is supported** iff $D_{r,i} > 0$ in at least 7 of 8 panels in each of the three replicates, $\text{median}_i D_{r,i} > u_r$ for each replicate, and every model passed the learning gate; it is rejected iff all gates passed and a condition failed; otherwise it is inconclusive. H2 uses $\Delta_{\text{geom}} = \rho_s(d_A, d_{B32}) - \rho_s(d_A, d_{B1})$ over all pairs, and again over pairs with the same w_{32}, decided by sign consistency; distances on beliefs are Hellinger. Pair correlations are never significance tests.

4.4 Preregistration and frozen protocols

The pilot's hypotheses, measures and rules were committed in `PREREGISTRATION.md` at the end of stage S2, before any training; secondary analyses were registered blind in `PREREGISTRATION_X4.md`; every operational detail of S7/S8 was fixed before any S7 result (D-047, D-048). A change to a closed decision required the owner at checkpoint H3; the one such change was the addition of H3′ before any P result (D-037). Each follow-up protocol is frozen by SHA-256 in `agent-handoff/planning/responses/*-FROZEN-v1.md`; coordinator addenda may fix implementation footprint or bindings but may not change a gate, threshold, seed or decision rule. Every frozen protocol contains a threshold register, an outcome taxonomy with a licensed sentence for each outcome, and an explicit "never licensed" list. Three-valued outcomes (PASS / INCONCLUSIVE / FAIL_TECHNICAL, or the study-specific equivalents) prevent a technical failure from being read as a negative finding. Since C1′, protocols also include a *tolerance-attainability* table fixed before data (the "R13 lesson", §7), and since K4b an *oracle-scale check* that compares margins with what the oracle itself achieves (the "K4b lesson", §6).

4.5 Dual-track planning and independent review

From 14 October I had every planning task done twice, independently, under `agent-handoff/planning/PLANNING_PROTOCOL.md`: track X by Codex in a read-only sandbox and track L by a clean-context Claude subagent, with identical request files and base commits and no access to each other's output. The coordinator wrote a merge document recording agreements, conflicts, the decision on each and what each track caught that the other missed. A third, result-blind LLM review (Codex, or a clean-context Claude reviewer when Codex was unavailable) had to return its required changes, and a confirmation pass had to return CONFIRM_FOR_FREEZE before any freeze. For example, C1 went through 13, then 7, then 4 required corrections; X6S1 through 14 required changes, including a verification of the empirical-Bernstein proof. The merge of the C1 tracks found that half of one track's "random controls" exactly reproduced the primary intervention (`agent-handoff/MORNING_REPORT_20261015_EN.md`).

4.6 Independent test authors

For critical modules the tests are written from a written API contract (`API.md`) by a separate LLM author who never reads the implementation, and they are committed before the module enters the repository; X6S1 used two independent authors (A and B). Corrections to an author's tests are made by the same author or a clean-context successor that reads only the governing ruling and the test, never module code. The implementer never writes the acceptance tests. The coordinator independently reruns the full test suite at each code freeze (e.g. 4,288 passed, 0 failed, at the X6S1 freeze H; `agent-handoff/jobs/0086/review.md`).

4.7 Metered compute

Every scientific process runs under a meter that places the process tree in a Windows Job Object and records child CPU including exited descendants, meter CPU and a 0.1 s tail allowance per run into a store-local `cpu_ledger.json`. CPU is *measure-only* in the later studies (no CPU ceiling; a wall limit and memory limits guard against runaway jobs). D-2a revealed that its per-model supervisor had watched the virtual-environment launcher, not the training interpreter, so per-process CPU read about 0; the job-level meter was unaffected (`agent-handoff/jobs/0033/review.md`). A repair (commit `0c341aa`) moved accounting to the whole process tree, and X6S1's preflight includes a check (RUNTIME-ACCT) that a deliberately over-limit child is killed with its training interpreter.

4.8 Registries against data leakage

A fresh test set is fresh only if none of its histories were ever generated, read or used for fitting in an earlier stage. From C1′ onwards, each study builds an *exposure registry* of digests of every history that any earlier lineage could have exposed (C1′ registry v1: 3.36 million digests; K4b K-REG2: 6,024,521; X6S1: about 19.7 million new plus K-REG2), and separation gates refuse a fresh pool with any collision. Reservations (`agent-handoff/RES-K-REG1`, `RES-K-REG2`) forbid any other history-generating job while a registry window is open. Pytest scratch directories inside the project root once made a registry inventory incomplete; scratch now lives outside the root (`X:/Downloads/topo-scratch/<job>`).

4.9 Sealing until the decision stage

Quantities that could influence later choices are written to `sealed/` with their hashes and released only at the decision stage or at a stop. In K4b the validation means stayed sealed until the fresh result was recorded; in X6 stage 1 the learnability-gate statistics, the learning curves and the headroom diagnostics are sealed, and the gate record releases only PASS/FAIL/UNRESOLVED and the decision (`X6S1-FROZEN-v1.md` §4.7, §9 S9, §12 item 11). Any model prediction on test histories before E3 is defined as leakage. This document follows the same rule: no X6S1 model-performance number was reported in earlier versions, and the numbers in §10.2.1 come only from the decision-stage report written by E3.9 after unsealing (`x6-stage1-store/stages/E3z/report.json`).

4.10 The technical-correction rules

A documented implementation or environment defect may be corrected by a coordinator ruling that names the defect, the correction commit, the affected artifacts and the evidence of independence from the outcome, with *at most two correction jobs per defect*. The failed attempt moves append-only to `superseded/<job>-a<k>/`; unaffected artifacts must regenerate bit-identically. A failure that shows a protocol statement to be wrong is not a defect: it stops the study and returns to planning. Any post-data change of a threshold or tolerance needs the owner. Environment reruns that reproduce identical values do not count as corrections, and final outcomes are never rerun.

5 The pilot: exits E0–E4

5.1 E0: the pipeline recovers known topology

Question. Does the persistent-homology measurement detect a surface hole, as opposed to graph cycles? **Design.** Gate G-A with thresholds fixed before S1 (D-016): correct Betti numbers; $S_{\mathrm{cyl}} \geq 0.20$ and $S_{\mathrm{sq}} < 0.10$ on d_M; synthetic circle versus disk clouds ($M = 512$ in $\mathbb{R}^{128}$) classified correctly in at least 19 of 20 draws. **Result.** $S = 0.299$ (cylinder) and 0.037 (square) on d_M; 0.231

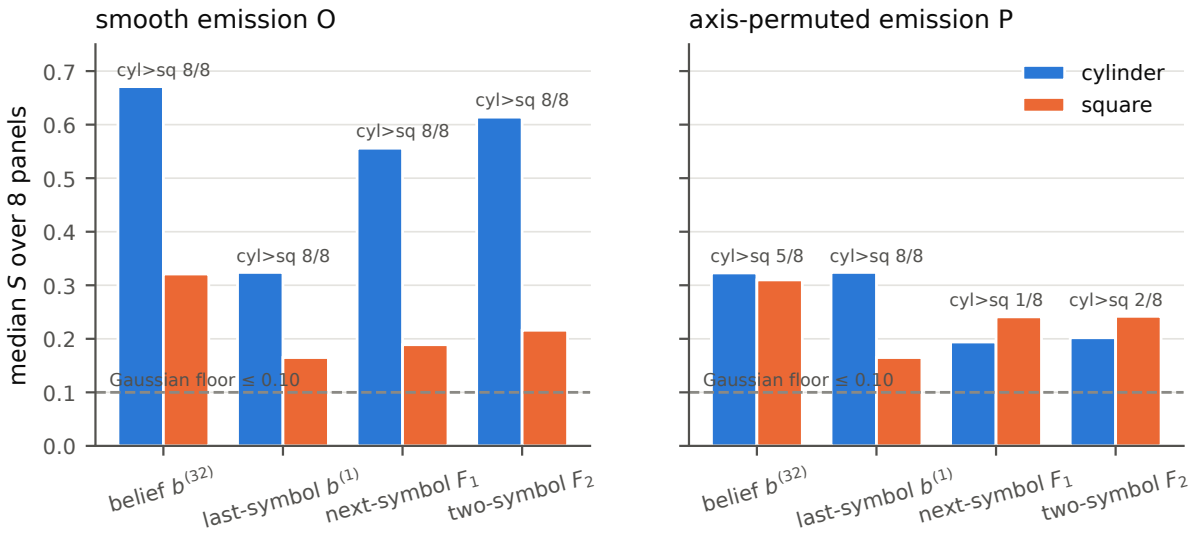


Figure 3: E1: median S over eight panels in the exact (unlearned) spaces, replicate 1729, $\sigma = 1.25$, Hellinger distances; labels give the number of panels with $S_{\text{cyl}} > S_{\text{sq}}$. Source: `README.md`, section "Earlier exit (E1, no model)".

and 0.067 on the graph metric, although the transition graphs have 133 and 121 independent cycles; synthetic calibration 20/20 (mean circle S 0.460, maximum disk S 0.045). G-A passed (`reports/E0/summary.md`). **Lesson.** Graph cycles are not surface holes; the measurement targets the latter.

5.2 E1: the topological difference exists in principle

Question. Do the model-free spaces separate the worlds? **Result.** Under O every space separates the worlds in 8/8 panels, at 3–8 times the Gaussian floor; the full belief separates most strongly (0.671 vs 0.321). Under P the exact belief barely separates (5/8 panels), and the future spaces reverse (Figure 3). The last-symbol posterior and the emission field are identical under O and P, because P permutes states. **Interpretation.** RQ1 is decidable under O, and history strengthens the expected signature. But H3 ("the structure survives under P") became *one-sided* before any P model was trained: a failure under P would be expected even of the ideal predictor. This reading was registered before the P runs.

5.3 E2: the main experiment under O

Design. Six runs (2 worlds × 3 seeds) plus one N model, 13–16 CPU minutes each. **What ran.** All six passed the learning gate (captured 0.985–0.986 for the cylinder and 0.992 for the square; regret 0.0018–0.0039 nats against $G_C = 0.2649$ / 0.2360). **Result.** F1 **supported**: $D > 0$ in 8/8 panels in each replicate, median $D = 0.158$, 0.176, 0.173 against $u_r = 0.097$, 0.084, 0.084 (Figure 4a). At initialisation the worlds are indistinguishable ($S \approx 0.06$); after training the cylinder reaches $S \approx 0.31$ and the square ≈ 0.13. H2 is sign-consistent: $\Delta_{\text{geom}} > 0$ in 48/48 panels over all pairs (medians 0.087–0.127), and among same-w_{32} pairs, where R1 is constant, the activation distances track $b^{(32)}$ with ρ_s 0.935–0.979. The N model sits at $H(W) = \ln 36$ (NLL 3.5835, half-width 0.0002): no leakage. In the secondary X4 analysis the F1 rule holds in all six views, including the readout-null component V4 (`reports/E2/summary.md`). **The R1 limitation.** The trained embedding of the last symbol alone reproduces 0.89, 0.47 and 0.54 of the model's median D; the model-free mixing baselines R2 and R3 stay at ≤ 0.20 (Figure 4b). F1 alone therefore does not show that history integration produces the hole.

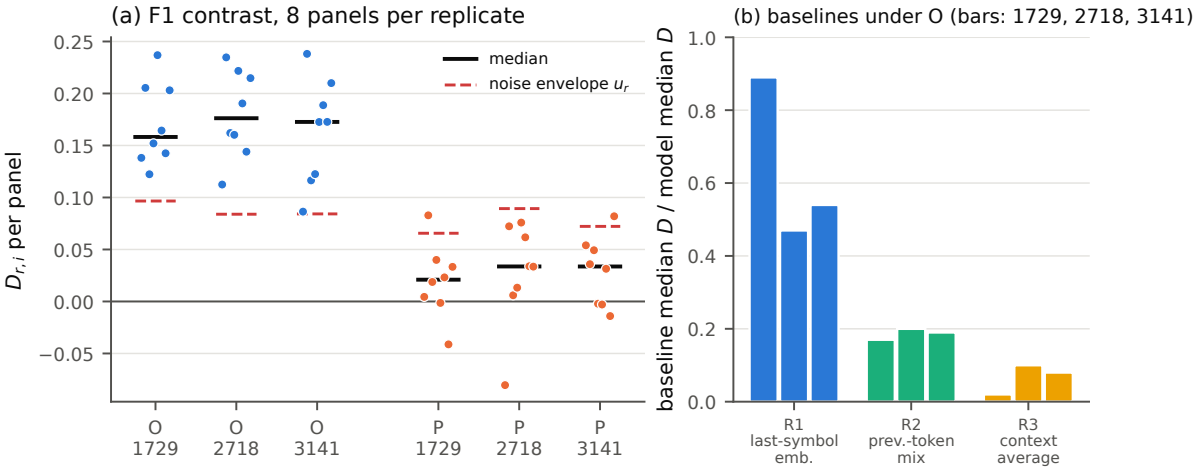


Figure 4: (a) F1 contrast $D_{r,i}$ per panel under O (S4) and P (S5), with the median and the noise envelope u_r per replicate. (b) Fraction of the model's median D that each representation baseline reproduces under O. Sources: `reports/he_summary/data/d.json` (per-panel D, extracted read-only from the artifact store by `reports/he_summary/extract_data.py`); u_r and ratios from `reports/E2/summary.md` and `reports/E3/summary.md`.

5.4 E3: the permuted control P and training dynamics

P. All six P runs passed the learning gate (captured 0.874–0.914). The F1 rule was **rejected** (median D 0.021–0.034 below u_r 0.066–0.089; 6/8 and 5/8 positive panels in two replicates), which, as preregistered, is uninformative. The R1 contrast collapses under P (replicate 1729: 0.140 $\rightarrow$ −0.035), so R1's O contrast is not static emission geometry; but whether it reflects history integration or learned symbol-level transition statistics remains open. **H3′ supported**: under P, $\rho_s(d_A, d_{B32})$ rises from 0.06–0.11 at initialisation to 0.70 and exceeds $\max(\text{R2}, \text{R3}) = 0.39$ in 48/48 run-panels. *Learned representations can align with Bayesian belief geometry even when the registered H_1 contrast is not supported*; an H_1 contrast is expected only when the belief geometry itself carries the hole.

H4 (timing). With t_h the first checkpoint where regret $< G_C$ and t_s the first where $S_t - S_0 > u_r$, all three cylinder runs under O had $t_h = 100$ and $t_s = 50$: **H4 rejected** (D-034; Figure 5). The early rise of S also occurs in the square, so it is not cylinder-specific. Exploratorily, the cylinder–square contrast on panel 1 grows mainly between steps 100 and 200 (+0.02 to +0.07 at step 50, +0.18 to +0.23 at step 200); the rule was written on raw S, not on the contrast, so this observation does not alter the verdict. A layer sweep shows the O contrast already at the embedding (≈ 0.105) and largest after the final LayerNorm (≈ 0.28), with P near zero at every hook (Figure 6).

5.5 E4: controls, probes, circular coordinates, sensitivity and the clean rerun

Controls. Two further N models sit at $\ln 36$ within 2×10^{-5}. The U models learn the visible-state dynamics (captured$_U$ 0.999) and also form an x-aligned circle (agreement 0.601), showing that symbol-level transition statistics alone can produce such a circle. **Probes.** Ridge probes fitted on dev panels decode $b^{(32)}$ from activations with test R^2 0.829 under O—no better than from the exact next-symbol law F_1 (0.838), far above initialisation (0.483) and the one-hot last symbol (0.421); the shuffled-label control is near zero (Figure 7a). The preregistered pattern (activations above all three competitors) held in 1/6 runs per condition. Under O the selected ridge λ was the smallest grid value in every run, so R^2 may be slightly underestimated; the grid was frozen. **Circular coordinates.** From the leading H_1 cocycle (over $\mathbb{F}_{47}$, at mid-lifetime)

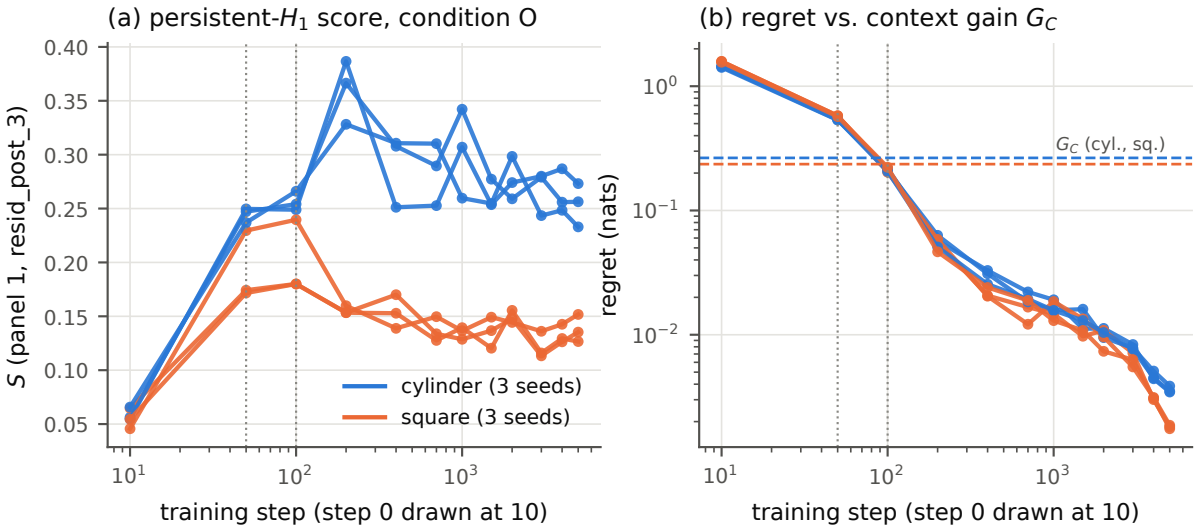


Figure 5: Training dynamics under O on panel 1: (a) the H_1 score S and (b) regret (log scale) at the twelve checkpoints, three seeds per world; dotted lines mark steps 50 and 100, dashed lines G_C. Sources: `reports/he_summary/data/dyn.json` (read-only extraction from the store, `reports/he_summary/extract_data.py`); G_C from `reports/E2/summary.md`.

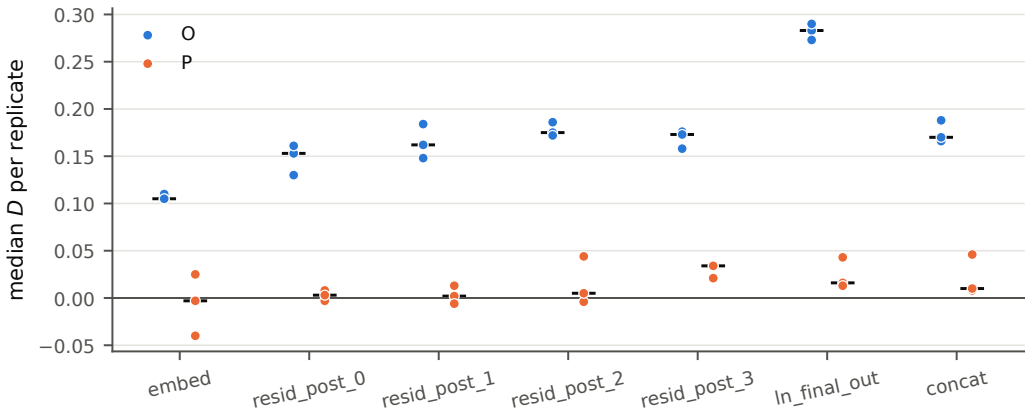


Figure 6: Layer sweep (descriptive): median D per replicate (dots) and its median (bar) for every hook, under O and P. Source: `reports/E3/summary.md`, table "Layer sweep".

a circular coordinate φ is computed and compared with the latent angle $\theta = 2\pi x(z_{32})/12$ by $A = \max(|\overline{e^{i(\varphi-\theta)}}|, |\overline{e^{i(\varphi+\theta)}}|)$. Cylinder models under O reach 0.884–0.892 (initialisation 0.06–0.12; non-periodic y 0.10; square 0.21–0.30; exact belief 0.915; Figure 7b). This identifies *which* loop F1 measures; it is privileged-evaluation and descriptive. A later diagnostic noted that the construction can silently give off-giant-component points an arbitrary offset; at most one point per panel was affected (`agent-handoff/jobs/0023/report.md`). **Sensitivity.** The verdicts are unchanged under the chord metric, dev-fitted PCA-32 and 256-history panels, and H3′ also under $\sqrt{\text{JSD}}$ (`reports/E4/summary.md` §6). **Clean rerun (S8).** A fresh clone with an empty store reran S0–S3 including one full training: all 307 artifacts shared with the original store were reproduced bitwise, all six gate verdicts matched, store integrity (785/785) and report reproduction (9/9 byte-identical) passed. Gate G-H nevertheless **failed**: one timing record of the original run did not exist, because the code that writes it was added after that run. The verdict stands as FAIL (D-049).

Lessons of the pilot. (i) A matched control must be evaluated on the oracle before training; without E1, the H3 rejection would have been misread. (ii) Alignment with belief geometry and a

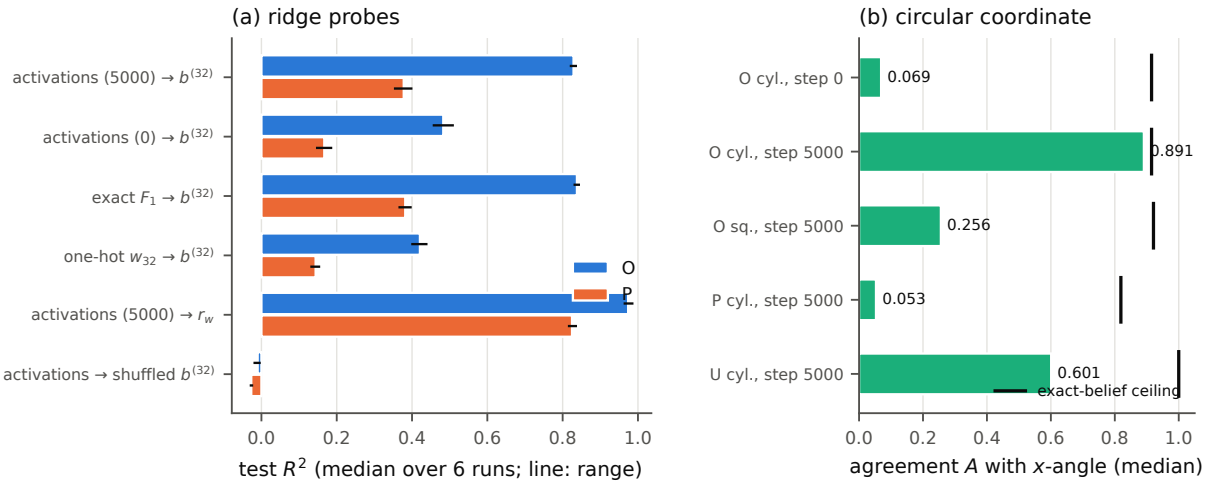


Figure 7: E4 (S7): (a) ridge-probe test R^2, median over six runs per condition with the range; (b) agreement of the circular coordinate with the latent x angle, with the exact-belief ceiling. Source: `reports/E4/summary.md` §4 and §5.

topological contrast are different properties. (iii) The wording of a rule determines what it can find: H4 was written on S, the interesting timing was in the contrast. (iv) "Bitwise reproducible" and "gate passed" are not the same statement when a gate is defined over a complete store.

6 Origin of the circular structure (charter priority 1)

6.1 Circular-baseline audit

Question. Does the learned circle exceed what the last symbol permits? **Design.** Descriptive, on the already-unblinded E4 panels (D-056). **Result.** In 24/24 panels the model's agreement (medians 0.884–0.892) exceeds an optimistic last-symbol bound (0.817) and the best available baseline (0.805): BEYOND_SYMBOL_AND_MIXING (D-057). **Interpretation.** This closes the missing last-symbol comparison of E4 descriptively; it is not independent confirmation and not evidence of long-history origin.

6.2 K8 history-origin cohort and the oracle profile

K8. A preregistered cohort aimed to compare the learned phase with information beyond the last 8 symbols. Its oracle headroom gate failed: $B_8 = 0.00925$ with a one-sided 95% upper endpoint 0.00949, below the 0.01 margin (`x4-history-feas-store/GATE_REPORT.json`). **Oracle profile (job 0014, exploratory).** For the first-harmonic target $m_K(h) = \mathbb{E}[c(z_{32}) \mid w_{33-K:32}]$, $c(z) = (\cos 2\pi x/12, \sin 2\pi x/12)$,

$$B_K = \mathbb{E}\,\|m_{32} - m_K\|, \qquad K \in \{1, 2, 4, 8, 16, 32\}, \tag{6}$$

was 0.369, 0.198, 0.067, 0.0092, 0.00018 and 0 on calibration (validation nearly identical; Figure 8a). All bootstrap intervals were recomputed independently. The coordinator's review corrected the worker's report: "only at $K \leq 4$" means only among evaluated grid points, and "this world cannot support memory beyond about 8" is not an accepted conclusion (`agent-handoff/jobs/0014/review.md`). **Lesson.** The circular posterior converges quickly with suffix length, so any history-origin experiment needs a small K.

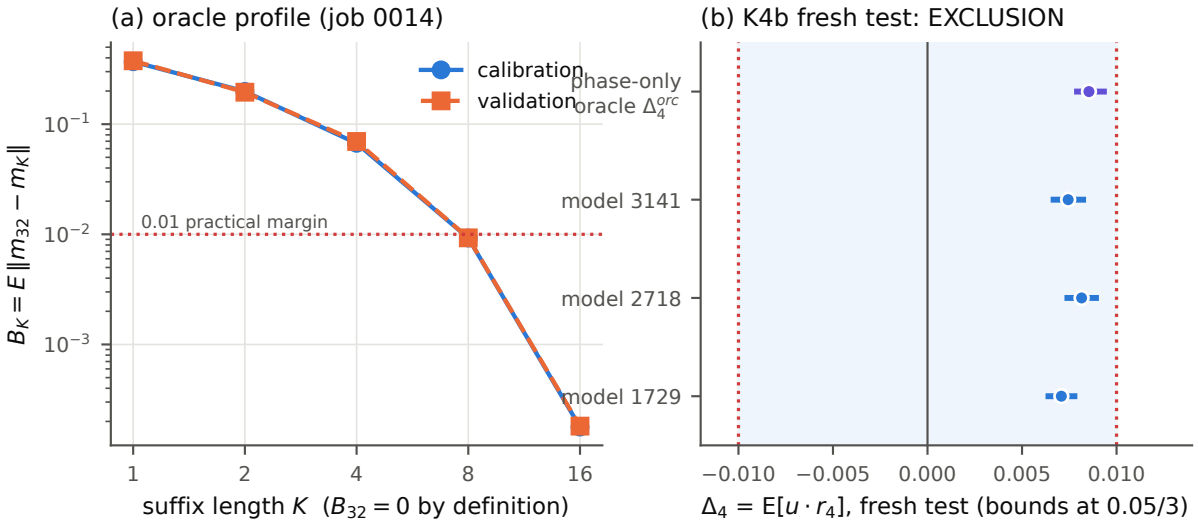


Figure 8: (a) Oracle profile B_K for the first-harmonic circular posterior (calibration and validation); $B_{32} = 0$ by definition. (b) K4b fresh-test result: per model the nominal one-sided bounds $[L, U]$ (rank-84 and rank-4,917 bootstrap order statistics) and point estimate of Δ_4, the EXCLUSION margin ± 0.01 (dotted), and the phase-only oracle comparator Δ_4^{orc}. Sources: `agent-handoff/jobs/0014/review.md`; `agent-handoff/jobs/0057/review.md`.

6.3 K4: the frozen chart rule fails

Design. K4 moved to $K = 4$ with a frozen circular-chart construction (the midpoint scale of the longest H_1 bar, landmark graph, integer lift). **What ran.** Identity and the exact process/oracle gates passed (moment errors $\leq 8 \times 10^{-16}$). **Result.** INCONCLUSIVE at chart_calibration: at the frozen scale the landmark graph of model 1729 was disconnected; by rule, no retry, no refit and no scoring (`agent-handoff/jobs/0018/review.md`). Because C1 depended on the K4 chart by its own freeze, C1 was blocked at the same moment. A geometry-only diagnostic (CHART-DIAG-FROZEN-v1) then found that the disconnection was a single isolated smallest-norm point and that the largest connected component held at least 99.8% of landmarks in all six cases (GO). I approved a new rule (the largest component, gated at $\geq 95\%$) for future studies only; K4 stayed closed.

6.4 K4b: does the learned phase covary with information beyond four symbols?

Question. In the three E4 cylinder-O models (step 5000, `resid_post_3`, position 32), does the circular coordinate $u_j(h) = (\cos\varphi_j, \sin\varphi_j)$ built by the new chart rule on fresh landmarks covary with $r_4 = m_{32} - m_4$, the part of the first-harmonic belief that the last four observations do not determine? **Estimand.** $\Delta_4 = \mathbb{E}[u \cdot r_4]$. By the tower property $\Delta_4 = 0$ for every suffix-only coordinate. Rows without chart support contribute the interval $[-\|r_4\|, +\|r_4\|]$, so $[\mathbb{E}X^-, \mathbb{E}X^+]$ is exactly the identified set over all completions. **Decision.** Per model, SUPPORT iff $L(\mathbb{E}X^-) - F > 0.01$; EXCLUDE iff $U(\mathbb{E}X^+) + F < 0.01$ and $L(\mathbb{E}X^-) - F > -0.01$, with one-sided bounds at 0.05/3 and a numerical margin $F = \max(B_{\text{num}}, 10^{-9}) + Z_{\text{bound}}$ that makes the claim concern the documented law (`K4B-FROZEN-v1.md` §5.5, §8). **What ran.** E1 (calibration/validation) needed three technical corrections, each within the two-per-defect rule: a dictionary passed to `Fraction` (R25), a byte-order mark and a lazy sequence in the registry handoff (R26), and a missing reader for the store's own superseded digests (R28); no outcome-bearing quantity existed at any failure point. Validation then passed every gate (support 3072/3072 in every model, B4 lower bound 0.0636) and fixed $N = 3072$ test rows. **Result.** On the fresh test, **EXCLUDE in all three models**: $[L, U] = [0.00640, 0.00777]$, $[0.00740, 0.00891]$, $[0.00667, 0.00823]$ (Figure 8b).

Suffix-only nulls were near zero, as designed.

Interpretation and caveat. The licensed sentence is that, for this chart rule, site and process, a covariance of either sign at or beyond 0.01 is excluded. But the phase-only oracle comparator itself achieves only $\Delta_4^{orc} = 0.00854$ [0.00791, 0.00933], below the margin, and the models reach 83–95% of it; all lower bounds are positive. EXCLUSION therefore means only that the covariance is small in absolute terms; it does not show that the learned phase lacks beyond-four-symbol information. The protocol registered no "positive but below margin" status, so no stronger claim is licensed (`agent-handoff/jobs/0057/review.md`). **Lesson.** Margins must be checked against the oracle's own achievable value before freezing; X6S1 adopted this as its oracle-scale check.

6.5 X2a stage 0: component identity beyond short windows (oracle only)

Question. In a process whose components have identical single-symbol statistics, how much component information is available from short windows versus the full history, and is the full history more than a product of block likelihoods? **Design.** Two 12×12 tori with the pilot walk; component B emits A's field pulled back by the Dehn twist $\phi(x, y) = (x + y \bmod 12, y)$; exact enumeration for windows up to 4 and 2^{20} validated Monte Carlo sequences otherwise (`X2A-FROZEN-v1.md`). **Result.** GO_P1_NONADDITIVE: the last symbol carries no component information; window information rises from 0.002 nats (2 symbols) to 0.067 (8) and 0.325 (full history); full-history Bayes accuracy exceeds the 8-window accuracy by 0.191 (99% interval [0.190, 0.192]) and a four-block likelihood product by 0.0567 [0.0562, 0.0571], above the 0.03 threshold; a local-noise control was nearly additive (0.008) (`agent-handoff/jobs/0025/review.md`). **Interpretation.** GO licenses *designing* a trained stage only, about component identity rather than the circular coordinate; short windows are not blind. It motivated the search for a construction where short windows are *exactly* blind, which became X6.

7 Causal use (charter priority 2)

7.1 X4-F: erasing the circular coordinate

Design. Erase the direction carrying the coordinate m_{32} at the final site and compare the damage to prediction with random erasers matched in damage. **Result.** The targeted eraser removed linear recoverability, but a fixed RBF decoder still recovered the coordinate at 0.998 (LINEAR_COMPONENT_ONLY), and 0 of 10,000 random erasers fell inside the frozen damage-matching window (FAMILY_UNMATCHED). The causal endpoint was not run: INCONCLUSIVE (D-063; `docs/proposals/X4F_COORDINATOR_REVIEW_ADDENDUM.md`). **Lesson.** The circular information is nonlinearly and redundantly encoded; linear erasure cannot test its use.

7.2 C1 and C1′: crossed counterfactual state forcing

Question. At `resid_post_3`, position 32, of the E4 cylinder-O models, where only the final LayerNorm and unembedding follow, does the readout map a change of the learned chart phase into a change of the predictive distribution in the direction an exact counterfactual oracle prescribes? Because the unembedding has rank at most 35, the residual-to-output Jacobian has a kernel of dimension at least 93, so the test is not trivial (`C1-FROZEN-v1.md` §1.1). **Design.** Donors $H_{\pm}$ shift the oracle state along the circle; a Frisch–Waugh–Lovell phase map fitted on calibration rows defines the forced state; 32 Haar-random whitened 2-frames form the matched control family; gates bound damage to the decoded last symbol $\hat{w}$ and the decoded y-marginal (M2), support (M3), registration of the counterfactual, a positive control, and a numerical certificate of every oracle row. C1 was

blocked by the K4 chart failure; C1′ replaced the K4 dependency with the largest-component chart rule (`C1PRIME-DELTA-FROZEN-v1.md`, 33 anchored edits). **What ran.** The oracle stage S3 first failed technically (an integer/string key mismatch, ruling R11), then passed every gate on bit-identical histories: the exact belief predicts the forced-state effect with a one-sided lower bound of 1.12 nats of mean gain, far above 0.01 (`agent-handoff/jobs/0041/review.md`). Calibration S4 first failed at M1, a ridge-equivalence self-check; ruling R13 relaxed that tolerance after S3 data existed, and the rerun reached M2. **Result.** INCONCLUSIVE/FAMILY_UNMATCHED at M2-cal in all three models. The primary intervention flipped the decoded w_{32} on 29–39% of rows ($D_w(T) = 0.311$, 0.389, 0.286 against a ceiling of 0.02) and shifted the decoded y-marginal by 2.5–3.2 times its calibration variance (ceiling 0.05), while the matched controls barely moved $\hat{w}$ (< 0.001); only 6, 0 and 4 of 32 frames matched, against 28 required. The coordinator found no implementation defect: the gap is a property of the fitted direction, not of displacement size (`agent-handoff/jobs/0044/review.md`). **Governance flag.** Ruling R13 changed a tolerance after data existed, without my prior approval, in tension with the protocol's own §19.3 and §19.8. The M2 values do not depend on R13 (charts and earlier gate values are bit-identical), and under a strict reading the outcome would have been FAIL_TECHNICAL at M1, which licenses the same "assay infeasible as specified". I ratified R13 in addendum 10. Since then every protocol carries a tolerance-attainability table fixed before data.

Interpretation for RQ3. No causal endpoint has been scored. Both failures tell me about the instruments and nothing about use: the information is nonlinear, and the obvious linear intervention direction is entangled with the last-symbol readout. A successor design (C2, constraining the intervention to preserve the last-symbol and y readouts and the pre-LayerNorm mean and variance) was the first of two independent plans. After a comparison, a merge and three rounds of result-blind review, I froze the successor as C2-FROZEN-v1 (`docs/proposals/planning/responses/C2-FROZEN-v1.md`). It asks whether the position-32 readout uses a registered circle code, by exact interchange of natural donor histories at the LayerNorm output and at layer 2; its implementation is under way.

8 Necessity and prediction horizons (charter priority 3)

8.1 X4 Phase B and B2: horizon-2 witnesses in the pilot world

Question. Are there reachable beliefs that agree in F_1 but differ in F_2, so that an ideal predictor must represent more than the next-symbol law? **Result.** The ambient tangent ranks allow it ($Q_1 = 35$, $Q_2 = 137$), but the four old pools were undercovered (49–66 eligible anchors against 128 required): INCONCLUSIVE (D-052). Four fresh pools (159/141/159/156 anchors) gave **zero** witnesses: a scoped NO-GO for the practical horizon gap at those thresholds and that resolution (D-054/055). It is not a universal sufficiency theorem; Phase C was not run.

8.2 D-1: an orientability distinction needed only at horizon 2 (oracle)

Design. A minimal 4×3 seam band, cylinder or Möbius, with a hidden world bit Z (prior $\frac{1}{2}$); five tokens $\sigma_0, \sigma_2, \sigma_3, A, B$. Theorems in the frozen protocol: at every decision position ($w_t = \sigma_3$) the next-symbol law is identical in both worlds; at horizon 2 the world-conditional laws differ only in four entries. **What ran.** Exact rational enumeration of all 37,190 reachable histories up to length 8, a float oracle agreeing to 4×10^{-15}, and an 80-digit Decimal reference on 650 histories. **Result.** PASS: every pre-stated exact value was reproduced, including the posterior-isolation pair (identical location beliefs, world posteriors $\frac{1}{4}$ vs $\frac{3}{4}$, identical next-symbol law, two-step TV $\frac{1}{100}$) (`agent-handoff/jobs/0020/review.md`). **Licence.** An existence result about an orientability distinction (equal Betti numbers) at reachable histories; nothing about the square and cylinder, β_1, or trained models.

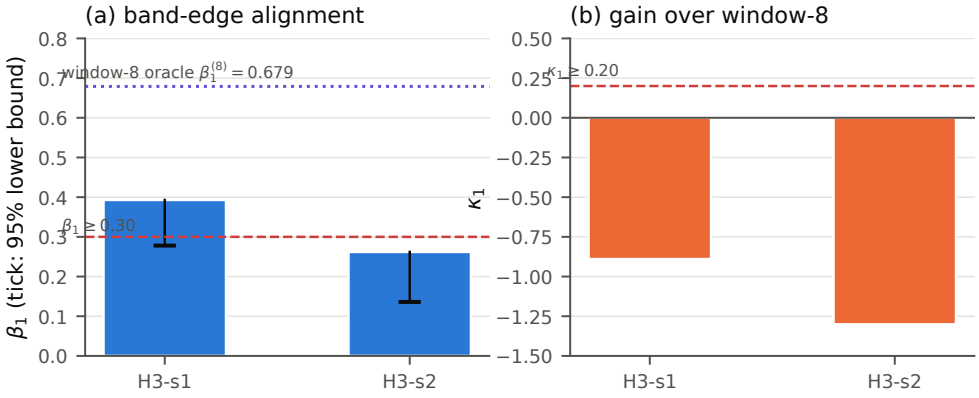


Figure 9: D-2a learnability gate on validation: (a) band-edge alignment β_1 with its 95% lower bound against the threshold 0.30 and the exact window-8 value; (b) the normalised gain κ_1 against 0.20. Source: `agent-handoff/jobs/0033/review.md`.

8.3 D-2a: do trained transformers carry the world posterior?

Question. At decision positions, how much of $\pi(h) = P(Z = M \mid h)$ is recoverable from the activation, and how does it depend on the training objective horizon $H \in \{1, 2, 3\}$ and on the in-context decision position $t \in \{16, 28\}$? **Design.** 4 paired seeds $\times$ 3 horizons plus 2 null models (14 models, 10,000 updates); two Bonferroni-controlled confirmatory contrasts; a 2-model learnability gate first. At band-edge positions the oracle contrast $d_i = 2(a/4)(\pi - \frac{1}{2})(\delta_M - \delta_C)$ is compared with the model's $r_i = [\hat{q}(A) - \hat{q}(B)] - [Q_1^{1/2}(A) - Q_1^{1/2}(B)]$ by

$$\beta_1 = \frac{\sum_i r_i d_i}{\sum_i d_i^2}, \qquad \kappa_1 = \frac{\beta_1 - \beta_1^{(8)}}{1 - \beta_1^{(8)}}, \tag{7}$$

and learned(m) requires $\beta_1 \geq 0.30$, its lower bound ≥ 0.15, $\kappa_1 \geq 0.20$ and a positive paired lower bound of $\beta_1 - \beta_1^{(8)}$ (`D2A-FROZEN-v1.md` §6). **Result.** NOT_LEARNED: both H3 gate models are defined FAILs (β_1 0.393 and 0.262, κ_1 -0.890 and -1.301), below the exact window-8 predictor's $\beta_1^{(8)} = 0.679$ (Figure 9); the remaining 12 models were not trained (`agent-handoff/jobs/0033/review.md`). Dev cross-entropy was flat at about 1.12 from step 1,000. **Licence.** "This recipe failed the registered learnability gate"—a scoped feasibility result that says nothing negative about representations.

9 Secondary programme X5: geometry and learning speed

Question (H5). Does a metric-coupled "conductance" kernel a020 (horizontal edge lengths $1 + 0.2\cos(2\pi y/11)$, conductances $1/l^2$) delay a grid-defined regret-threshold event by at least 250 steps relative to the flat kernel a0, in each of three paired models, on the same cylinder and emission? **Design.** 27-point step grid; PASS when mean regret $\leq 5\%$ of G_C at three consecutive grid points; paired bootstrap bounds (`docs/proposals/X5_G1_TRAINING_PROTOCOL_V4.md`). **Result.** Paired delays 250, 0 and 0 steps; one pair's upper bound is $0 < 250$, so H5 is **REJECTED** on dev, with no censoring (`agent-handoff/jobs/0013/review.md`); X5 was closed without a fresh test (D-081). **Interpretation.** The kernel change is confounded with movement speed (the feasibility gate showed a large speed difference), so this is not a curvature null. A mean-rate-matched successor is a listed but unscheduled option.

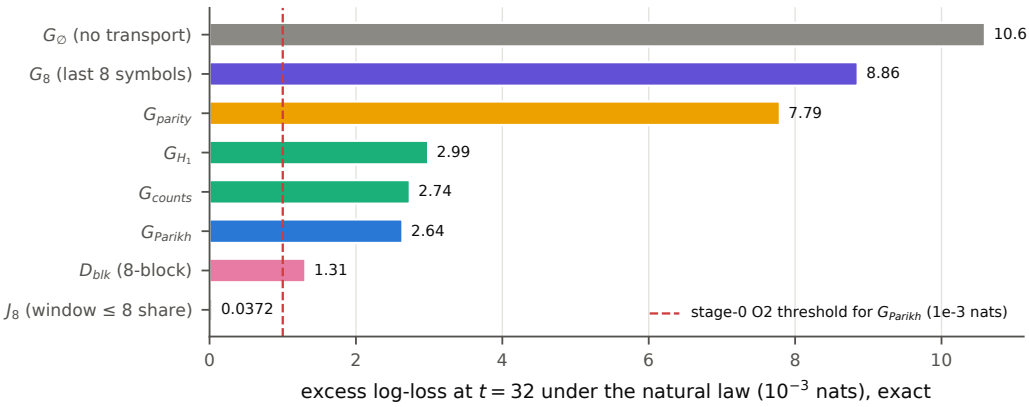


Figure 10: X6 stage 0: exact natural excess log-loss at $t = 32$ of the Bayes predictor restricted to each commutative statistic (no transport; last 8 symbols; parity; H_1 class; unsigned counts; Parikh vector), the 8-block comparator D_{blk}, and the share J_8 of the gap carried by windows of at most 8 symbols; dashed line: the O2 threshold. Every function of a statistic has at least that statistic's excess. Source: the certified stage-0 values quoted in `agent-handoff/planning/responses/X6S1-FROZEN-v1.md` §2.1, from `agent-handoff/jobs/0066/review.md` and `x6-stage0-oracle-store/run/report.json`.

10 Secondary programme X6: non-abelian transport

10.1 Stage 0: order-dependence certified in the oracle

Construction. A 4×4 torus with the pilot's lazy walk (each move and stay with probability $1/5$). Each vertex emits its own token except an anchor $a_\star = (2, 2)$, which emits a cue revealing a hidden fibre value $x \in \{0, 1, 2\}$. Crossing the x-seam applies the transposition $a = (01)$ to the fibre, crossing the y-seam applies $b = (12)$; every face is flat except one corner face with a 3-cycle flux, and the fibre is redrawn at rate $\eta = 1/50$ (a 48-state lift). For a segment γ since the last cue, the holonomy is $\rho(\gamma) = g_{e_n} \circ \cdots \circ g_{e_1}$ and the Parikh vector $\kappa(\gamma) = (k_{x^+}, k_{x^-}, k_{y^+}, k_{y^-})$ counts directed seam crossings. Abelian controls AB (a on both seams) and ZC (3-cycles) have the same counts but no order effect (`X6-FROZEN-v1.md` §2). **Outcome criteria.** O1: on designed witness families the cue-normalised TV between members of different holonomy is ≥ 0.3 while every commutative competitor is constant; O2: the natural excess log-loss of the best Parikh predictor $G_{\text{Parikh}}(32) \geq 10^{-3}$ nats and the fraction M_{32} of histories with row-KL ≥ 0.01 is $\geq 1/32$, both as certified enclosures. **What ran.** Two technical corrections (a key accumulation in G5, ruling X7; a sign error in a G6 estimator, ruling X8), each confirmed by the next run. **Result.** GO_X6_NATURAL (`agent-handoff/jobs/0066/review.md`): witness TVs 0.851, 0.695, 0.524; exact zeros in AB, ZC and the controls; $G_{\text{Parikh}}(32) = 2.636 \times 10^{-3}$ nats (enclosure width $\sim 10^{-60}$) and $M_{32} = 0.0468$; no window of at most 7 symbols carries order information (exact). The ladder of commutative competitors is in Figure 10. **Licence.** Oracle facts about one designed process; a GO supports designing a trained stage, it does not authorise training. A descriptive note: the literal product-of-experts comparator has a transport-independent component (TRIV ≈ 19 standard errors above zero), so it is not used as an order signal in stage 1.

10.2 Stage 1: do trained transformers use crossing order?

Question. Do small causal transformers trained only by next-symbol prediction on natural NA histories make predictions that depend on the order of seam crossings, beyond the Parikh statistic and its coarsenings? The claim is behavioural; no mechanism is claimed (`X6S1-FROZEN-v1.md` §1).

Design (Figure 11). Arms NA, AB and ZC $\times$ seeds 1–4: 12 models, each trained for 30,000 updates on 2^{20} fresh histories with the pilot architecture and recipe; for each seed every arm starts

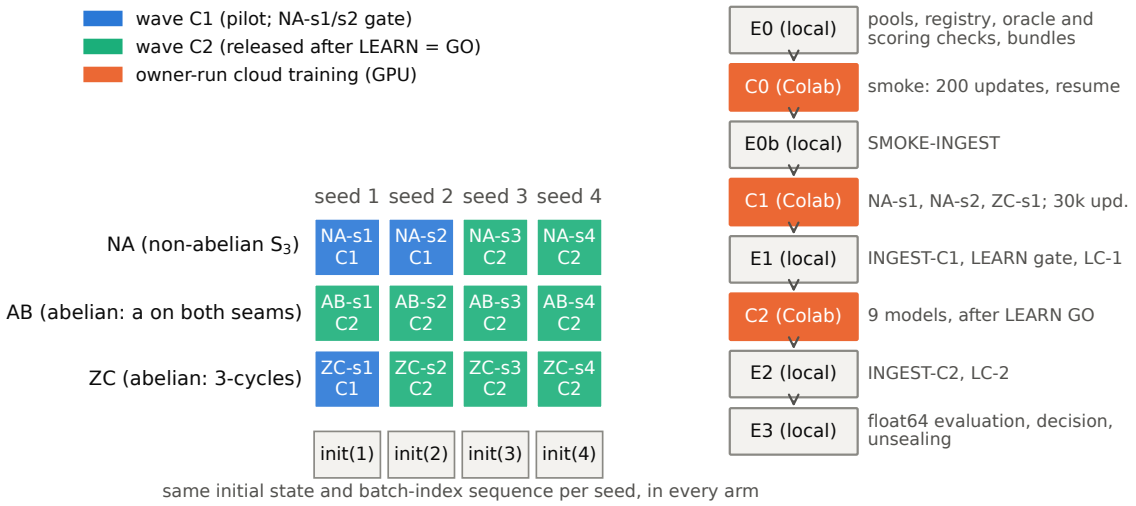


Figure 11: X6 stage 1 design: three arms × four seeds sharing four initial states, two training waves, and the alternation of local and cloud stages that I ran. Source: `agent-handoff/planning/responses/X6S1-FROZEN-v1.md` §4.1–§4.4, §5, §8; wave parameters (C0: 200 steps, $K = 100$; C1 and C2: 30,000 steps, $K = 1{,}000$) from `x6-stage1-store/stages/E0/frozen_C*.json` (bundle id, steps and K only), via `docs/research_summary/data/x6s1_status.json`.

from the same initial state and uses the same batch-index sequence. Training runs on the Colab free-tier GPU in resumable 1,000-update chunks (model, optimiser, scheduler and RNG state); every outcome-bearing evaluation runs locally on CPU in float64 from hashed weights. Waves: C0 (smoke), C1 (NA-s1, NA-s2, ZC-s1), C2 (the other nine, released only after the learnability gate says GO). Local stages: E0 (pools, registries, oracle and scoring checks, bundles), E0b (smoke ingest), E1 (ingest C1, learnability gate, learning curves), E2 (ingest C2, learning curves), E3 (evaluation, decision, unsealing).

Primary estimand. Each eligible test history h (a cue present, age $n \geq 8$, end vertex next to the anchor) is paired with an exact uniform draw h' from its own Parikh class (same prefix, cue, age, end vertex and directed crossing counts). With $\Delta\hat{q} = \hat{p}_c(h) - \hat{p}_c(h')$ and $\Delta q^* = q^*_c(h) - q^*_c(h')$ on the cue-normalised laws,

$$\beta_{\mathrm{ord}}(m) = \frac{\sum_i \langle \Delta\hat{q}_i, \Delta q^*_i \rangle}{\sum_i \|\Delta q^*_i\|^2}. \tag{8}$$

Every function of the Parikh key scores exactly 0 (the pair shares the key), every function of the cue-free segment—including every function of the last 8 symbols—scores 0 in expectation, and the oracle scores 1. The control-adjusted slope is $\tilde{\beta}(s) = \beta_{\mathrm{ord}}(\text{NA-}s) - \max\{\beta_{\mathrm{ord}}(\mathrm{init}(s)), \overline{\beta}_{\mathrm{ord}}(\mathrm{AB}), \overline{\beta}_{\mathrm{ord}}(\mathrm{ZC})\}$, and **C-ORD** is supported iff $\mathrm{mean}_s \tilde{\beta}(s) \geq 0.25$, its 97.5% bootstrap lower bound exceeds 0.05, $\tilde{\beta}(s) > 0$ in at least 3 of 4 seeds, and a tail diagnostic passes.

Log-loss claims and the empirical-Bernstein bound. For the quantised, ε-smoothed readout ($Q = 2^{20}$, $\varepsilon = 10^{-6}$) the per-row excess loss $X_i = \mathrm{KL}(q^*(H_i) \,\|\, p^Q(H_i))$ lies in $[0, B]$ with $B = 17$. With $n = 2m = 2^{21}$ rows, $\bar{X} = \frac{1}{n}\sum X_i$, $Z_k = (X_{2k-1} - X_{2k})^2/2$, $\bar{Z} = \frac{1}{m}\sum Z_k$, $L = \ln(2/\delta)$, $a = B\sqrt{L/m}$, $b = B^2 L/(12m)$ and $\sigma_{\mathrm{up}} = (a + \sqrt{a^2 + 4(\bar{Z} + b)})/2$, the protocol proves

$$P\Big(\mu \leq \bar{X} + \sigma_{\mathrm{up}}\sqrt{2L/n} + \frac{BL}{3n}\Big) \geq 1 - \delta \tag{9}$$

from a one-sided Bernstein moment bound and a paired U-statistic for the variance (`X6S1-FROZEN-v1.md` §7.3), and computes the bound outward in exact and directed-rounding arithmetic. **SHARP**(s) holds iff the bound at $\delta = 0.025/4$ is below the stage-0 lower endpoint of $G_{\mathrm{Parikh}}(32)$: then model NA-s is not a function of the Parikh statistic or any coarser one. A secondary

TARGET requires recovering 25% of that gap. Designed criteria use the bag- and bigram-matched AB-versus-BA family (3,090 pairs).

Learnability gate. Before any order statistic exists, an NA pilot model must have natural excess log-loss credibly below the exact window-8 gap $G_8(32) = 8.855 \times 10^{-3}$ nats on fresh validation rows (bootstrap upper bound at 97.5%). Passing means "better than every function of the last 8 symbols"; it does not mean order use. GO iff NA-s1 or NA-s2 passes. Only statuses and the decision are released; the statistics are sealed.

What has run (to 20 October 2026, ∼13:35Z; Figure 13).

- *Implementation* (jobs 0069–0086): API contract and two independent test authors, eleven modules within line limits set by coordinator addenda 2–5 (footprint only; no scientific change), a coordinator full suite of 4,288 passed at code freeze H.
- *Authorisation*: my addendum 17 authorised the frozen protocol (SHA-256 2A909ACE…), a local budget of about 43 CPU-h (forecast range 23–70), and cloud use of about 1–4 GPU-h.
- *E0, attempt 1* (job 0087): AUTH-ADMIT PASS, S0 INVALID; nothing scientific ran. Two defects: a constant path one level short of its value (D-0087-1) and S0's lineage check refusing the AUTH-ADMIT record written by the same run (D-0087-2). Corrected in job 0088.
- *E0, attempt 2* (job 0089): all ten gates PASS (AUTH-ADMIT, S0, PF, G0-ORC, SCORE, G0-POS, REG, G0-FEAS, HEADROOM-D, CLOUD-PREP), then AUTH-VERIFY PASS; bundles C0, C1, C2 built.
- *C0 and E0b* (jobs 0090–0092): I ran the smoke in Colab. The first ingest was INVALID at CLOUD-PROV because a hash pasted with a line break produced a cell that did not compile but was still recorded; the provenance check treated every recorded source as executed (D-0090-1). The fix skips sources that do not compile, which cannot have executed anything; the second ingest on the same snapshot PASSED. An open defect (D-0090-2) records that no bundle re-issue mechanism exists for the retrain route; it is not needed unless a retrain occurs.
- *C1 and E1* (job 0093): INGEST-C1 PASS (all four integrity gates for the three models), **LEARN PASS with the released decision GO (NA-s1 PASS, NA-s2 PASS)**, LC-1 PASS.
- *C2 and E2* (job 0094): I ran C2; S0, INGEST-C2 (all four integrity gates for the nine C2 models) and LC-2 PASSED, so all 12 models are admitted. The coordinator accepted E2 and scheduled the E3.A waves and then E3z.
- *E3* (jobs 0095, 0097–0099): the first evaluation attempt (0095) was INVALID because every model chunk exceeded the 3 GiB per-process supervisor limit (S6; defect D-0095-1, a full-pool vectorised forward peaking at about 3.0 GiB against a forecast of 1.2 GB). The correction evaluates in row blocks of 2^{16} with byte-identical outputs (peak 1.11 GiB) and was re-frozen; the regenerated initialisation chunk was bit-identical. E3.A then completed all 16 predictor chunks (0097, 0098), and E3z (0099) passed every step E3.2–E3.9 and unsealed.

Total local compute was 79.55 CPU-h, about 21 CPU-h of it in the failed attempt (Figure 14b); I extended the budget beyond the approved 23–70 CPU-h (my addendum 18). Cloud training took one Colab session per wave, about 10 s (C0), 24 min (C1) and 69 min (C2) by the session records.

10.2.1 Results (decision stage E3z)

All numbers in this subsection are from `x6-stage1-store/stages/E3z/report.json` (sha256 `90be2081...`), written by E3.9 after unsealing, and the coordinator's review `agent-handoff/jobs/0099/review.md`. Figure 12 summarises them.

Outcome. INCONCLUSIVE by rule 9 of §7.9, with licence "descriptive only" and qualifiers CONTROL_LIMITED(designed, s1), CONTROLS_LEARNED(AB 4/4, ZC 4/4), LADDER(NA-s1..s4: parity), DESIGNED_UNRESOLVED(1) and LENGTH_PROFILE. Every validity gate passed, all 12 models were admitted and learned (LEARN GO; LEARN-TEST PASS for all twelve), the natural placebos were clean, the tail diagnostic passed and precision was adequate in every seed.

Natural order use. On the informative natural partner pairs at $t = 32$ (the oracle has $\beta_{\mathrm{ord}} = 1$, the Parikh-class predictor 0), the four NA models reach $\beta_{\mathrm{ord}} = 0.047,\ 0.090,\ 0.072,\ 0.049$, the AB controls 0.001–0.004, the ZC controls 0.001–0.008 and the initialisations about 0. The per-seed contrasts are $C_s = 0.044,\ 0.088,\ 0.064,\ 0.045$, each with a 95% interval far from zero, and the mean is $\tilde{\beta} = 0.061$ with $\mathrm{LB}_{0.975} = 0.0595$. The registered C-ORD claim requires a mean of at least 0.25, so C-ORD is *not supported*; lowN ($\tilde{\beta}(s) < 0.10$) holds in all four seeds and orderN in none.

Designed order use. On the designed W2 witnesses the NA models' regret is $R_{W2} \approx 0.080$–0.087, above the bar $(3/4)\Delta^{\mathrm{lo}}_{W2} \approx 0.0458$, so ordD fails in seeds 2–4. Seed 1 is unresolved because the commutative control ZC-s1 responded to the designed W2 contrast (placebo $\beta_{W2} = 0.209$; other controls ≤ 0.006 in magnitude). Descriptively, the ZC controls reach a W2 regret similar to the NA models' (0.077–0.094) while the AB controls stay near 0.20, so the designed regret does not separate order-sensitive from commutative models. SHARP and TARGET were certified for no NA model, and the finest commutative ladder rung the NA models beat is parity.

Why rule 9. Rule 8 (ORDER_NOT_DETECTED) needs all four designed flags resolved. Every other condition of rule 8 holds; the single designed placebo failure of ZC-s1 leaves ordD(s1) unresolved, and the rules are applied as registered, without reclassification.

Reading. Within the licence: next-token training on natural non-abelian transport histories produced a weak but reproducible sensitivity to the order of seam crossings, present in every seed and absent from the commutative controls, at about one sixteenth of the oracle's order information. It falls far short of the order use the study was designed to detect, and the designed probes do not show it. Whether more training, larger models or a different curriculum would close the gap is a new question.

11 Engineering and process

11.1 Coordinator, workers and the mailbox

I organised execution around a separate git repository, `agent-handoff`, holding the protocol (`PROTOCOL.md`), one directory per job (`task.md`, `status.json`, `report.md`, `review.md`), the request and control files, planning requests and responses, owner decisions (immutable files plus numbered addenda) and the coordinator's state (`COORDINATOR_STATE.md`, newest "CURSOR" first). Codex acted as coordinator until 14 October, followed by a chain of Claude coordinator sessions (c351b245, 0cb12c0f, e4713512, 75dcd488, dcecc9ff). Each takeover wrote an assumption record, because release verifiers bind the issuer's identity; one takeover race was resolved by having the late session stand down. Workers were headless sessions launched per job, with context caps that forced rotation to fresh sessions; a transfer record registers each new worker identity. 94 jobs had been created by 20 October. Two operational hazards recurred: a shared subscription hitting usage limits (handled by waiting, never by switching accounts), and permission classifiers blocking some actions (never bypassed). I approved and installed an automatic-resume scheduler.

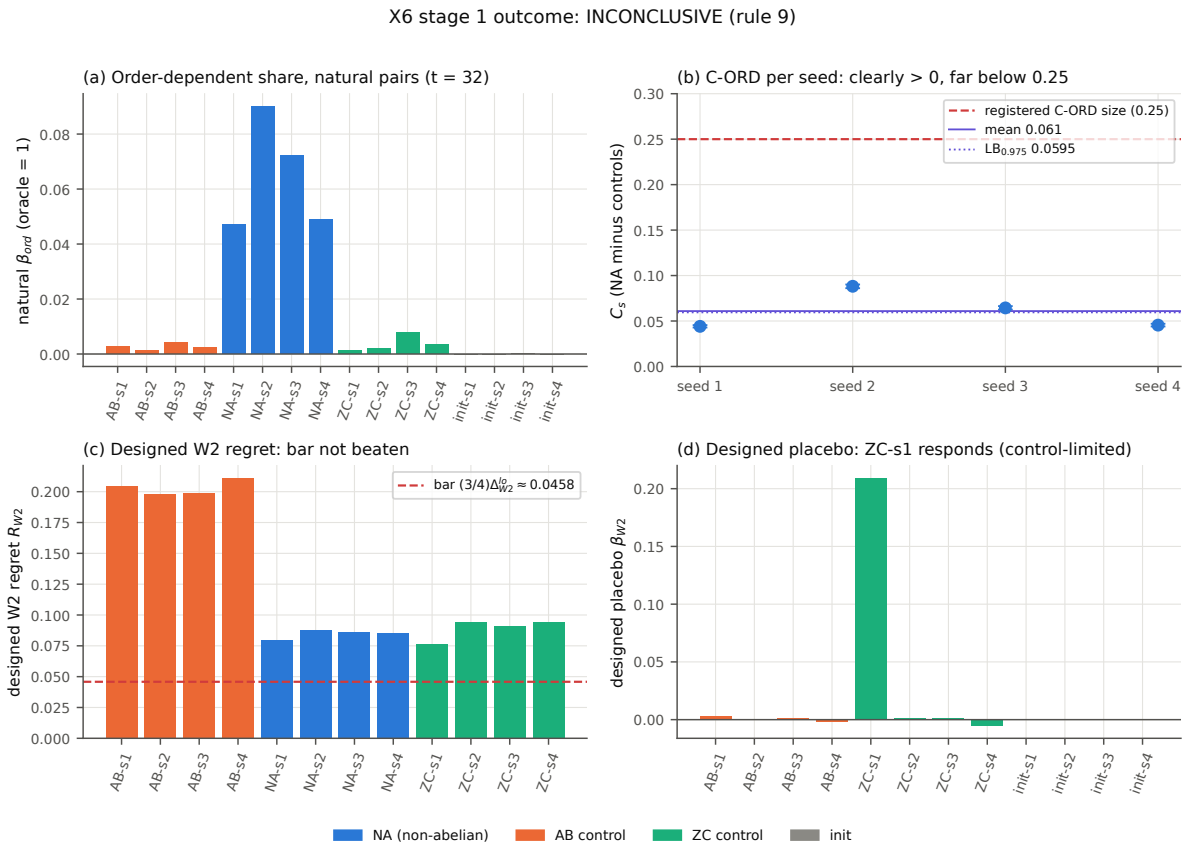


Figure 12: X6 stage 1 decision (E3z). (a) Natural β_{ord} per predictor on the informative $t = 32$ partner pairs. (b) Per-seed C-ORD contrast C_s with 95% intervals, the mean and $\text{LB}_{0.975}$, against the registered effect size 0.25. (c) Designed W2 regret against the bar $(3/4)\Delta^{\text{lo}}_{W2}$. (d) Designed placebo β_{W2} of the controls (ZC-s1 fails). Source: `x6-stage1-store/stages/E3z/report.json` via the snapshot `docs/research_summary/data/x6s1_result.json`.

11.2 The cloud training pipeline and its integrity gates

X6S1 is the first study trained on a GPU. The coordinator builds deterministic bundles (code archive, data, initial states, a ticket with seeds, steps and checkpoint interval, expected batch and RNG digests for every chunk) and releases a bundle hash to me, and I run a fixed notebook in the browser; only one cell (the bundle hash) may be edited. Every chunk records its batch digest and PCG64 state digest; the export carries a manifest whose hash I relay. Locally, four gates run per model before any evaluation: WEIGHTS-HASH (manifest, inventory and canonical weights digests), CLOUD-PROV (every session and execution record matches the permitted cell templates, commit, protocol and bundle), DATA-ID (data, initial state, per-chunk batch and RNG digests), and TRAIN-COMPLETE (a linked chain of exactly one record per checkpoint). An admission lock then fixes the model's only admissible chain. GPU training is not claimed to be reproducible; data, batch sequence, initial state and every evaluation are.

11.3 Defects found and corrected

The correction route was exercised many times, always before any outcome-bearing quantity existed or with documented independence from it. Examples: K4's 2π phase edge (job 0017); D-1's module split under the 400-line rule (S6 stop, addendum 2); C1′'s int/str key mismatch (R11) and ridge tolerance (R13, discussed above); K4b's `Fraction(dict)` (R25), BOM and lazy sequence (R26), fixture roots inside the project (R27) and missing reader (R28); X6 stage 0's key

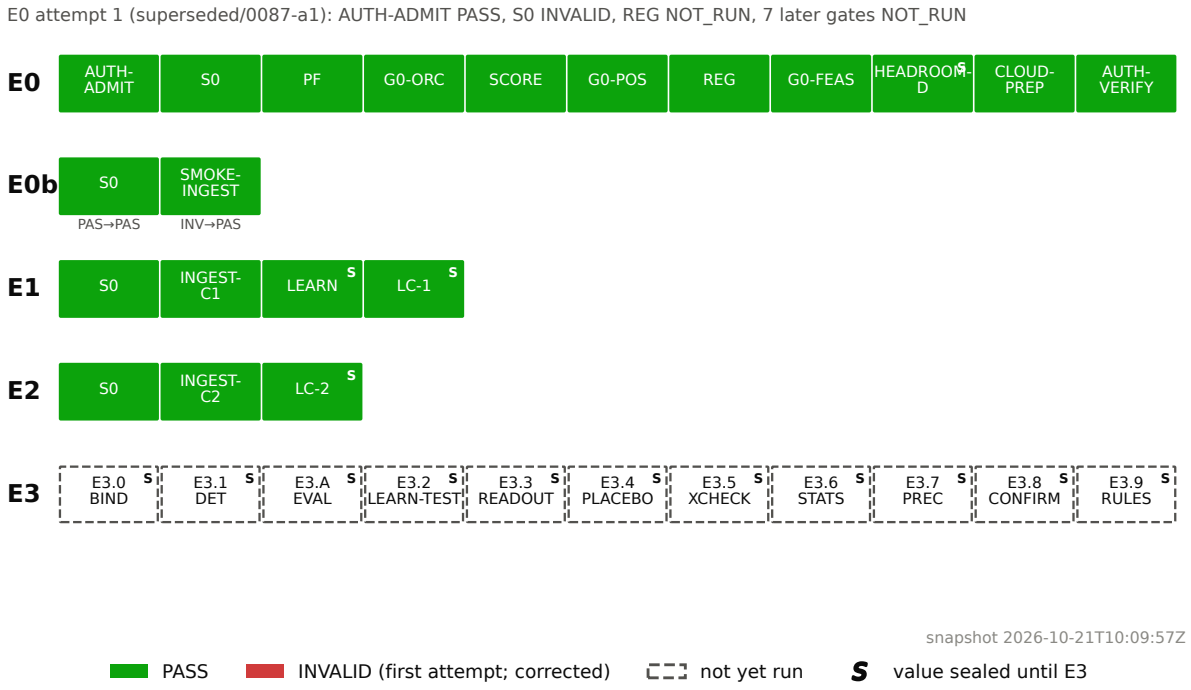


Figure 13: X6 stage 1 gate pipeline with statuses. Filled: PASS; outlined dashed: not yet run; "S": the gate's values are sealed until E3; arrows below a box give the status history over attempts. Statuses and gate names only. Source: `x6-stage1-store/GATES.jsonl` and `superseded/0087-a1/GATES.jsonl` (fields gate, status, utc only; the values field is never read), via the snapshot `docs/research_summary/data/x6s1_status.json`; stage order from `X6S1-FROZEN-v1.md` §8.

accumulation (X7) and estimator sign (X8); X6S1's constant path and E0 lineage (D-0087-1/2) and the non-compiling pasted cell (D-0090-1). Two classes of defect were caught only by real runs: those where tests exercised synthetic inputs but not the real output of a reused function (R25), and those that depend on the live environment (the pasted cell). Both led to added checks.

12 Compute

Figure 14 summarises local compute from the store ledgers. Including every metered run (preflight test suites among them), the follow-up studies used about 17.6 CPU-hours in the ledgers shown; the 17 pilot trainings took 4.4 wall-clock hours on a six-core laptop CPU (13–21 minutes each). The largest items are the two training studies (X5 G1, six models; D-2a, two models), X6 stage 0's exact dynamic programme, and X6 stage 1 so far.

13 Current status and next steps

- **X6 stage 1** is decided: INCONCLUSIVE (rule 9; §10.2.1). No study is running. Natural follow-ups for me to plan include the ZC-s1 designed-placebo response, the gap between the observed order sensitivity ($\tilde{\beta} \approx 0.06$) and the registered size (0.25) under longer training or larger models, and the open retrain mechanism (D-0090-2).
- **Causal use.** The C2 constrained-intervention draft awaits dual-track planning and review.
- **Other options** recorded by the coordinator and by me include an independent replication of the E4 O models (R1), a mean-rate-matched X5 successor, a D-2 learnability redesign, a

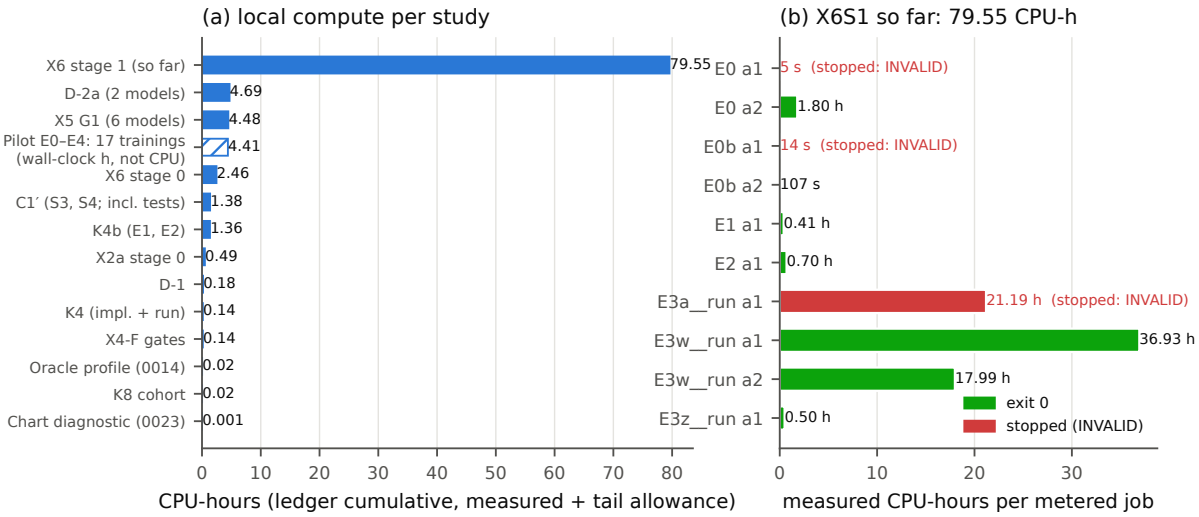


Figure 14: (a) Cumulative CPU-hours per study from each store's `cpu_ledger.json` (measured plus tail allowance; includes metered test suites); the hatched bar is the summed wall-clock time of the 17 pilot trainings, which predate the meter. (b) X6 stage 1 measured CPU per metered job and attempt. Sources: `<store>/cpu_ledger.json` for the stores named in `make_figures.py` (`LEDGERS`), snapshotted in `docs/research_summary/data/compute_ledgers.json`; `x6-stage1-store/cpu_ledger.json` (accounting fields only) via `data/x6s1_status.json`; `artifacts/MetricRecords/*/extra.json` (`wall_time_s`).

follow-up on the K4b margin and comparator scale, and the remaining tasks of a seven-task project reorganisation. None is authorised.

- **Records out of date.** `state/PROGRESS.md` and the README's "Current follow-up status" date from 17 October and still describe K4b job 0051 as running; K4b completed with EXCLUSION (job 0057), X6 stage 0 completed with GO_X6_NATURAL (job 0066), and X6 stage 1 started. The mailbox `STATUS_EN.md` is a historical snapshot from job 0010. This document reflects the mailbox reviews, which are the more recent records.

14 Threats to validity and limitations

1. **No causal claim.** All pilot evidence is representational; both intervention designs stopped at gates.
2. **One toy setting.** One small architecture and recipe, synthetic HMMs on tiny lattices, a few seeds (three in the pilot, at most four per arm). Nothing is claimed about natural language or large models, other architectures, or populations of seeds.
3. **Statistics.** Pilot decisions use sign consistency and a data-derived envelope, not significance tests; panels are partitions within runs. Most later bounds are nominal bootstrap bounds; only X6S1's log-loss claims carry a finite-sample guarantee.
4. **Measurement choices.** Vietoris–Rips over $\mathbb{F}_2$, $M = 512$, the longest bar as S; four one-factor sensitivity variants. Circular coordinates use one cocycle and can give disconnected points arbitrary offsets.
5. **Exposure.** The circular audit, C1′ and K4b reuse the three E4 models, whose test panels had been unblinded; K4b and C1′ drew fresh histories protected by registries, but the models are not fresh.
6. **Margins and controls.** K4b's margin sits above the oracle's own value; the P control is one-sided; X5's kernel is confounded with speed.

7. **Reproducibility scope.** One machine and environment; one of 17 pilot trainings was retrained; G-H formally failed; GPU training in X6S1 is not reproducible by design.
8. **Process.** One post-data tolerance change (R13) occurred without my prior approval and I ratified it afterwards. Records written by agents were reviewed by other agents; I approved decision points but did not audit every computation. The text of this summary was produced with LLM assistance.

15 Conclusion

Under controlled conditions, next-token training on a hidden walk produces a learned circle that reflects the world's topology, aligns with Bayesian belief geometry, and exceeds what the last symbol permits. Where it comes from is only partly known: the last-symbol embedding carries much of it, beliefs converge after a few symbols, and the beyond-four-symbol covariance of the learned phase is below a margin that even the oracle's phase does not reach. Whether the model uses it is unknown. The follow-up programme mostly produced well-defined stops rather than findings, and the stops taught something about one-sided controls, nonlinear encodings, non-specific interventions, fragile chart rules and margins set without oracle checks. X6 stage 1 is designed to avoid each of these failure modes on a process where order provably matters. It reached its decision with every validity gate passed and found weak but reproducible natural order sensitivity in every seed, far below the registered effect size, so its preregistered verdict is INCONCLUSIVE and not a claim of order use.

A Glossary

Term	Meaning
O, P, N, U	Emission conditions: smooth Gaussian; axis-permuted; uniform; one symbol per state
$b^{(32)}$, $b^{(1)}$	Exact belief after 32 symbols; posterior from the last symbol alone
F_1, F_2	Exact next-symbol law; joint law of the next two symbols
G_C, regret, captured	Context gain; mean KL from the oracle to the model; $1 - \text{regret}/G_C$
S, $D_{r,i}$, u_r	Normalised longest finite H_1 bar; double contrast; noise envelope
Δ_{geom}	Spearman alignment with $b^{(32)}$ minus alignment with $b^{(1)}$
R0–R3	Baselines: initialisation; last-symbol embedding; previous-token mixing; context average
V0–V5	X4 views: raw; LN-normalised; final LN; readout-visible; readout-null; softmax
m_K, B_K, r_4	Circular posterior mean from the last K symbols; $\mathbb{E}\|m_{32} - m_K\|$; $m_{32} - m_4$
Δ_4	K4b estimand $\mathbb{E}[u \cdot r_4]$ for the learned chart phase u
Chart	Circular coordinate built from an H_1 cocycle on landmarks (LCC rule: largest component $\geq 95\%$)
Holonomy, local system	Composition of fibre permutations along a path; an assignment of permutations to edges, flat on a face if their product around it is the identity
Parikh vector	Counts of each directed seam crossing, ignoring order
$G_S(32)$	Excess log-loss at $t = 32$ of the best predictor that sees only statistic S
β_{ord}, C-ORD	Within-Parikh-class order slope; the primary X6S1 order claim
SHARP, TARGET	X6S1 log-loss claims below $G_{\text{Parikh}}(32)$, and below 75% of it
Gate	A check that must pass before a stage's products are interpreted
Registry	Set of digests of every previously exposed history, used to certify fresh pools
Sealed	Written with a hash and released only at the decision stage or a stop
Technical correction	Coordinator-ruled fix of a documented defect; at most two jobs per defect
Release (A/B, R1–R3)	Coordinator-signed permission for a specific stage of a study

B All stages, gates and outcomes

Stage / study	Date	Decisive gate(s)	Outcome	Record
E0 (S1)	10-13	G-A	PASS	D-017; reports/E0
E1 (S2)	10-13	G-B, G-C, G-D	PASS; P one-sided	README (E1)
E2 (S3–S4)	10-13	G-E, G-F; F1 rule; H2	F1 SUPPORTED; H2 consistent	reports/E2
E3 (S5–S6)	10-13	F1 rule under P; H3′; H4	H3 rejected (uninf.); H3′ SUPPORTED; H4 REJECTED	reports/E3
E4 (S7–S8)	10-13	G-G; G-H	G-G PASS; G-H FAIL	D-049; reports/E4

Stage / study	Date	Decisive gate(s)	Outcome	Record
X4 Phase B	10-13	coverage	INCONCLUSIVE	D-052
X4 Phase B2	10-14	witnesses	scoped NO-GO	D-054, D-055
Circular audit	10-14	descriptive	BEYOND_ SYMBOL_AND_ MIXING	D-057
X4-F	10-14	G3, G4	INCONCLUSIVE / FAMILY_ UNMATCHED	D-063
K8 cohort	10-14	oracle headroom	FAIL	x4-history-feas-store
X5 G1	10-14	H5 rule (dev)	REJECTED; closed (D-081)	job 0013; D-073
Oracle profile	10-14	exploratory	accepted, qualified	job 0014; D-074
K4	10-15	chart_calibration	INCONCLUSIVE	job 0018; D-078
C1	10-15	K4 chart dependency	blocked by freeze	job 0018 review
D-1	10-15	PF, S0, G0–G6	PASS	job 0020; D-080
Chart diagnostic	10-15	giant fraction ≥ 0.95	GO	job 0023
X2a stage 0	10-15	P3, P4, A3	GO_P1_ NONADDITIVE	job 0025; D-084
D-2a	10-15	learnability (stage 1)	NOT_LEARNED	job 0033
C1′ S3	10-16	G0, H1–H5, N0	PASS (after R11)	jobs 0039, 0041
C1′ S4	10-16	M1, M2-cal	INCONCLUSIVE / FAMILY_ UNMATCHED	jobs 0042, 0044
K4b E1	10-17	G0 … P1 (10 gates)	PASS (after R25, R26, R28)	jobs 0049–0055
K4b E2	10-17	H2-FRESH, T-PRE, ENDPOINT	EXCLUSION (3/3 models)	job 0057
X6 stage 0	10-18	PF, S0, G0–G7, D	GO_X6_NATURAL (after X7, X8)	jobs 0063–0066
X6S1 IMPL	10-20	gate 0	complete (freeze H)	job 0086
X6S1 E0	10-20	AUTH-ADMIT … CLOUD-PREP, AUTH-VERIFY	attempt 1 INVALID (S0); attempt 2 all PASS	jobs 0087, 0089
X6S1 E0b	10-20	SMOKE-INGEST	attempt 1 INVALID; attempt 2 PASS	jobs 0090, 0092
X6S1 E1	10-20	INGEST-C1, LEARN, LC-1	PASS; LEARN = GO	job 0093
X6S1 E2	10-20	INGEST-C2, LC-2	PASS; 12 models admitted	job 0094
X6S1 E3	—	E3.0–E3.9	all PASS; INCONCLUSIVE (rule 9)	0095–0099

C Sources

- Pilot: `README.md`; `docs/RESEARCH_SUMMARY.md`; `docs/RESEARCH_PLAN_EN.md`; `PREREGISTRATION.md`; `PREREGISTRATION_X4.md`; `reports/E0` to `reports/E4`; `state/DECISIONS.md`.
- Follow-ups: `docs/proposals/`; the frozen protocols in `agent-handoff/planning/responses/`; the job reviews and reports in `agent-handoff/jobs/`.

- Coordination: `agent-handoff/COORDINATOR_STATE.md`; `agent-handoff/MORNING_REPORT_20261015_EN.md`; `agent-handoff/NEXT_RESEARCH_AND_WORKFLOW_EN.md`.
- Figure data and their provenance: `docs/research_summary/make_figures.py` and `docs/research_summary/data/`.

Paths starting with `agent-handoff/` or naming a `*-store` directory are execution records kept next to the repository, not in it.

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